Fitting and choosing radiative-transfer emulators for the retrieval

Yitzchak Shmalo*

Abstract

Radiative-transfer emulators are fitted to a code's outputs but used through an inverse, the retrieval of surface reflectance, and here they are fitted and chosen on the retrieval error. On the EMIT table, networks trained on the square root of the retrieval weights floored at a hundredth of the median transmission lower the radiance error by 8% and the 95th percentiles of the retrieval error by 32 to 44% on every one of ten splits, and they lower every tail on every split at reflectances from 0.1 to 0.9 with each failed inversion counted as an infinite error; a kernel fitted on learned network features keeps the lowest retrieval tails on this table. On the libRadtran benchmark of Mazid and Rishe the same weights lower the radiance error from 0.91% to 0.11% and the retrieval tail from 4.6 to 0.35 points on every one of ten seeds, at a larger error in the spherical albedo, and on states beyond the training range of aerosol and water vapour they lower the root-mean-square error as well. Two failed evaluations of the code in the EMIT table make every weighted network worse than the plain one, and refitting a convex stack on the retrieval error lowers its tail by a quarter. The theory behind these results gives the retrieval error exactly in terms of the component errors and shows that the forward and retrieval rankings of two emulators differ by a change of measure with density proportional to t^2/q^4. With the transmission floored, training on the retrieval error is regression under a known covariate shift whose price in effective sample size can be computed before any training; without the floor the shift is not integrable.

1 Introduction

At each band the top-of-atmosphere radiance over a Lambertian surface of reflectance ρ is

$$L(\rho) = a + \frac{\rho t}{1 - \rho s},$$

with path radiance a, transmission t and spherical albedo s, and atmospheric correction recovers ρ from a measured radiance by inverting this relation [96]. Computing (a, t, s) with a radiative-transfer code such as MODTRAN [7], libRadtran [64, 33] or 6S [101] for every pixel is too slow, and the codes are replaced by emulators fitted to tables of their outputs [10, 81]. The emulators are fitted and compared by their forward error, the error in the emulated components or in the radiance they produce, but they are used through the inverse. An error in the emulated components enters the retrieved reflectance divided by the slope $\partial_\rho L = t/q^2$, $q = 1 - \rho s$, so that the same forward error costs little where the transmission is large and a great deal where it is small. On tables that reach into strong absorption the two criteria can rank emulators differently: on the EMIT atmospheric-correction table a convex stack of kernel and network emulators has the lowest radiance error of the emulators compared and a larger tail of the retrieval error than the best of its members [85].

*Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel. yitzchak.shmalo@gmail.com.

The question here is what an emulator should be fitted to, and chosen by, when it is used through an inverse. The experiments come first and the theory that accounts for them second.

Networks trained on the retrieval error show what the weights buy and what they cost (Section 5.3). On the EMIT table the floored weights at a tenth of the median transmission lower the tail of the retrieval error on the physical domain on every split, by nearly a quarter to more than a third on average, and leave the radiance error unchanged, and a rule fixed before the runs, which chooses the floor on the validation block, picks that floor on every split. At lower floors, which keep fewer effective entries, the weighted objective still falls but the radiance error grows, and at the lowest floor every tail with it; the objective at a low floor is attained best on held-out states by the fit one decade above it. Flattening the weights, $\bar{w}_\tau^{\kappa}$ with $0 < \kappa < 1$ [84], does better than any floor: at a hundredth of the median transmission and $\kappa = 1/2$ the network lowers the radiance error by 8% and the four tails by 32 to 44% on average, each on every split, and it attains the floored objectives on held-out states better than the networks fitted on them; only the albedo error grows, by less than half. Kernel ridge regression fitted on the floored weights, a reweighting with truncated weights as in [61], repeats the pattern of the floors (Section 5.5): at a tenth of the median transmission it lowers every physical tail on every split, by 9 to 15%, at 12% more radiance error, and at a thousandth its objective on held-out states falls to 0.08 of the plain fit's while the radiance error and two of the tails rise. Scored at reflectances from 0.1 to 0.9 with every failed inversion counted as an infinite error, the floored and flattened networks lower every tail on every split, and the kernel on learned features of [85] keeps the lowest retrieval tails of all the emulators at every reflectance (Table 2).

The table holds two failed evaluations of the code, with zero transmission in most bands. A floored weight gives such an entry the weight of u^{-2} ordinary ones, and with the two in the table every weighted network is worse on held-out states than the plain one, on nearly every split even on the weighted objective itself. On a libRadtran table, where the floored weights fall on one band of thirteen, no floor improves on the plain network in both the radiance error and the tails, and the same rule picks one that is worse in the radiance error and in the physical tails. On the published benchmark of [65], the same corpus split by its release into 35,000 training states, with the 6S coefficients as an input and the band as one, the flattened and floored weights lower the radiance error by factors of 8.5 and 6.9 and every retrieval tail by factors of 10 to 13 on every one of ten seeds; in the release's own metrics they cut the mean absolute error to 0.27 and 0.33 of its best model's and raise the root-mean-square error by a quarter, all of it from the spherical albedo. On states above the training range of the aerosol optical depth or the water-vapour column the same weights lower the root-mean-square error too, by 28 and 30% on every seed (Section 5.4).

Choosing a few numbers on the retrieval error costs much less than fitting on it. Refitting the weights of the convex stack on the first-order retrieval error, with the members fitted on the forward error, lowers the 95th percentile of the retrieval error on the physical domain below that of the component stack on 39 of 40 held-out halves of the EMIT table, by a quarter at the median, to the level of the kernel on learned features, the best single member (Section 5.6). Chosen on the cut criterion, the input length scales of a kernel regression move toward the water-vapour column and nearly halve its tail on the physical domain, at four and a half times its radiance error; at that radiance error, about the networks' own, the tail is a fifth of the plain network's (Section 5.7). The stopping epoch of a network, a choice among many close candidates, has little to give either way (Section 5.8).

The theory of Section 6 accounts for these results. The retrieval error has an exact closed form in the three component errors (Lemma 6.1). To first order it is the forward error at ρ divided by the slope, and over a range of reflectances it is a Mahalanobis norm of the component errors with a metric that depends on the state only through (t, s) and is positive definite once three reflectances

are used (Proposition 6.3). The forward and retrieval rankings of two emulators differ by a change of measure with density proportional to t^2/q^4 (Proposition 6.5), and on the EMIT table the convex stack's net excess of squared retrieval error lies in the tenth of the entries where that density is smallest (Section 5.1). With the transmission floored, training on the retrieval error is regression under a covariate shift whose likelihood ratio is known in closed form and bounded by the floor; without the floor the ratio is not integrable once the transmissions approach zero as those of the EMIT table do, so the floor defines the target rather than approximating one (Section 6.3). The rate that a fit certified in the weighted norm guarantees is sharp (Propositions 6.6 and 6.7), and the price of the weighting can be computed before any training: the effective sample size of the floored weights is of the order of the number of entries below the floor (Proposition 6.8), and a fraction $(2-\beta)/2$ of their weight lies below the floor (Corollary 6.9). On the EMIT table that weight sits in about thirty of the 285 bands, those of the water-vapour absorption, and it keeps most of the states (Section 5.2). The price of choosing among K candidates on a weighted criterion is set by the effective number of states rather than of entries (Proposition 6.11), and the retrieval weights keep three quarters of the validation states or more; every stack that no other improves on both counts minimizes one of a one-parameter family of mixed objectives (Proposition 6.12).

Outline. Section 2 reviews related work. Section 3 sets out the retrieval error and the weights used throughout, Section 4 the tables, the emulators and the scores, and Section 5 the experiments. Section 6 gives the theory: the exact retrieval error and its metric, the change of measure, what a weighted fit guarantees and what it costs, and the price of choosing on a weighted criterion. Section 7 discusses the results, and Appendix A holds the proofs.

2 Related work

Radiative transfer and its inversion. The relation $L = a+\rho t/(1-\rho s)$ for a Lambertian surface goes back to the modelling of satellite measurements of ground reflectance in [28]. Operational atmospheric correction evaluates it with a radiative-transfer code, 6S and its vector version [101, 51], libRadtran [64, 33] or MODTRAN [7], directly or through lookup tables, as in ATCOR [76] and in the algorithms reviewed in [34]. For imaging spectroscopy the retrieval is posed as optimal estimation [78]: ISOFIT inverts the surface and the atmosphere jointly for EMIT [96], and the uncertainty of the retrieved reflectance is quantified in [97]. The carbon-dioxide retrievals of OCO-2 invert a full-physics forward model in the same framework [17, 25, 20, 29], and fast principal-component approximations of the radiative transfer [50] have been judged by the errors they cause in these retrievals [87, 88]. The benchmark of Section 5.4 uses the thirteen bands of Sentinel-2 [27].

Emulators of radiative transfer. Networks have replaced parts of atmospheric models since the emulation of longwave radiation in a climate model [53], and more recently for gas optics and for the whole transfer calculation [99, 98]. Learned parameterizations of subgrid processes [74] are fitted offline and used inside the model, and accuracy offline does not guarantee a stable coupled simulation [9], a counterpart of the gap between forward and retrieval error studied here. In remote sensing, statistical emulators of radiative-transfer models are surveyed in [103], packaged in the toolbox of [77], used for global sensitivity analysis [102] and built on Gaussian processes [67, 13]. The sRTMnet emulator corrects a fast approximation toward MODTRAN inside ISOFIT [10]; Vicent Servera et al. [81] compare regression emulators of the MODTRAN transfer functions and evaluate the corrected reflectance; Lamminpää et al. [1] fit a Gaussian-process emulator of the OCO-2 forward model by cross-validated forward error and use its analytic Jacobians in the retrieval.

Surrogates of radiative transfer also serve inversions of exoplanet atmospheres [41], atmospheric sounding [82] and, for atmospheric transport, greenhouse-gas flux inversions [24]. The correction of a cheap code toward an expensive one is multi-fidelity modelling [46, 71, 70] in the tradition of the design and analysis of computer experiments [79]; the pKANrtm emulator of [65] fits the libRadtran residual of 6S with a Kolmogorov–Arnold network [60, 49]. The emulators compared in [85] and used here are networks, Matérn kernel regressions [90, 73] and their combinations. None of these fits is weighted by the retrieval.

Surrogates for inverse problems. In Bayesian inverse problems [44, 92] the effect of a surrogate on the solution has been treated as an approximation error in the likelihood [45], bounded through moments of the emulator error in the Hellinger distance between posteriors [93], and controlled by an L^2 error weighted by the posterior, which suggests where to place training points [40]. Surrogates refined where the posterior concentrates [57, 16, 35], corrected by the residual of the forward model [14], enriched with gradient information [80], or fitted on a reduced parameter space informed by the likelihood [21, 22, 107] all aim the fit at the inverse problem; the propagation of surrogate uncertainty into the posterior is studied in [75], and training data chosen to match the information needed downstream in [54]. Goal-oriented model reduction [11], goal-oriented inference [58, 59, 89] and goal-oriented surrogates for stochastic inversion [63] fit the reduced model to the quantity of interest. Derivative-informed training fits a network to the sensitivities as well as the values [23, 6], and derivative-informed neural operators [55, 56] accelerate Bayesian inversion [15]; [5] constrain the sensitivities of a neural operator for forward and inverse modelling. The weight here comes from the retrieval equation in closed form and acts on the three component errors jointly through a metric.

Training for the task. Fitting a model by the loss of the decision it feeds is decision-focused learning [26, 104, 62], with the predict-then-optimize loss of [31], its generalization bounds [3], learned surrogate losses [83] and decision-focused surrogate models in process engineering [38]. The retrieval is a decision of this kind, but its sensitivity to the emulator is known in closed form, so the task loss is a weighted forward loss and needs no differentiation through a solver.

Covariate shift and weights. Weighting by a known likelihood ratio is the classical answer to covariate shift [84, 95, 66], with importance-weighted cross-validation [94], estimated ratios [42] and a learning theory of the correction [18, 19]. It is needed when the model is misspecified and not when it is correctly specified [84]; for nonparametric models that are misspecified it is again needed for the best approximation under the target distribution [36], and the benefit of labels from the target depends on how singular the shift is [52]. For kernel ridge regression with an unbounded ratio of finite second moment the truncated reweighting of [61] is minimax optimal, and in general the estimation error of an importance-weighted fit is controlled by the second moment of the weights [19]. For overparameterized networks the effect of importance weights fades as training proceeds without regularization [12]. Truncating weights [43] and smoothing their tail [100] are the importance-sampling counterparts of the floor and of the flattening used here, and the effective sample size of a weighted average is Kish's [48], discussed at length in [32].

Choosing models and combining them. The kernel parameters of Section 5.7 are chosen on a criterion evaluated on held-out states; cross-validation [91, 2] and its closed form for ridge regression [37] are the usual criteria, and kernel flows [68, 39] learn kernel parameters by how much the kernel interpolant changes when half of the data are dropped. Stacking [106, 8] descends from

the combination of forecasts [4]. The networks are trained with Adam [47] and stopped on held-out data [72], in PyTorch [69].

3 The retrieval error and the weights

An emulator predicts $(\hat{a}, \hat{t}, \hat{s}) = (a + e_a, t + e_t, s + e_s)$ at each state and band, and the retrieval inverts the predicted relation at the true radiance,

$$\hat{\rho} = \frac{L - \hat{a}}{\hat{t} + \hat{s}(L - \hat{a})}, \qquad L = L(\rho).$$

Write $q = 1 - \rho s$, so that $\partial_\rho L = t/q^2$, and $e = (e_a, e_t, e_s)^\top$. To first order in the component errors (Corollary 6.2) the retrieval error is $\hat{\rho} - \rho \approx -d$ with

$$d = v_\rho^\top e, \qquad v_\rho = \Big(\frac{q^2}{t},\ \frac{\rho q}{t},\ \rho^2\Big)^\top,$$

and the forward error at ρ, $r = \hat{L}(\rho) - L(\rho)$ with $\hat{L}(\rho) = \hat{a} + \rho\hat{t}/(1 - \rho\hat{s})$, is $r \approx (t/q^2)\, d$: the retrieval error is the forward error divided by the slope. Hence $r^2 \approx g\, d^2$ with $g = t^2/q^4$, so that relative to the retrieval the forward error gives an entry of transmission t a weight of order t^2. In the retrieval, errors in the path radiance and in the transmission carry weights of order $1/t^2$, and the error in the spherical albedo a weight of order one.

The weights of this paper come from a bound on the retrieval error. The retrieval constrained to $[0, R]$, $\bar{\rho}$, satisfies $|\bar{\rho} - \rho| \le \min\{R, e_R/t\}$ with $e_R = |e_a| + R|e_t| + R^2 t|e_s|$ at every entry [85], and the transmission in the denominator is floored at $\tau = uT_{\text{train}}$, where T_{train} is the median positive transmission of the training block and u is called the floor. The *floored* weights are $w_\tau = \max\{t, \tau\}^{-2}$ and the *cut* weights $v_\tau = t^{-2}\mathbf{1}\{t > \tau\}$, which give no weight below the floor; normalized to mean one, $\bar{w}_\tau = w_\tau/\mathbb{E}[w_\tau]$. Two milder forms temper the floored weights: the *flattened* weights $\bar{w}_\tau^{\kappa}$ with $0 < \kappa < 1$, renormalized to mean one, the device proposed for covariate shift in [84], and the *mixed* weights $(1 - \alpha) + \alpha\bar{w}_\tau$ with $0 < \alpha < 1$, which keep part of the plain objective. A fit on n entries with weights w has the effective sample size $n_{\text{eff}} = n\,\mathbb{E}[w]^2/\mathbb{E}[w^2]$ [48], and n_{eff}/n is its effective fraction. Section 6 derives the exact retrieval error, the change of measure between the two rankings, and what these weights guarantee and cost.

4 Tables, emulators and scores

EMIT. The first table has 23,313 evaluations of a radiative-transfer code provided by the Jet Propulsion Laboratory as a lookup table for EMIT atmospheric correction. The input is the atmospheric and viewing state in $\mathbb{R}^6$ (aerosol optical depth, surface elevation, water-vapour column, relative azimuth, solar and view zenith angles), and the output is four spectral components on the 285-point EMIT grid from 381 to 2493 nm: path radiance $Y_1 = a$, direct and diffuse fluxes Y_2 and Y_3 with $t = Y_2 + Y_3$, and spherical albedo $Y_4 = s$ [85]. Each split draws a test block of 2,331 states and a validation block of 2,098 from a seeded permutation and fits on the remaining 18,884; the splits are the ten seeds 101–110 of [85]. The table holds two failed evaluations of the code, states 4011 and 7439. Below about 800 nm their transmission and albedo agree with those of their neighbours in the input space. Between 797 and 850 nm their fluxes fall to 10^{-4} of the neighbours' and their spherical albedo rises to 2; from about 1030 nm on both fluxes are exactly zero, in 197 of the 285 bands, and the albedo is at least one in 227 bands, exactly 2 in 107. No other state

has an albedo of one or more. Both are in the training block on eight splits; split 104 has one in its test block and split 105 one in its validation block. The networks of Section 5.3 are fitted with them and, removed from whichever block holds them, without them; the members of the stacks of Section 5.6, the kernels of Section 5.7 and the networks of Section 5.8 are fitted on the table as provided.

libRadtran. The second table is the libRadtran half of the corpus of [65] over the thirteen Sentinel-2 bands, restricted to the 9,722 states that carry all thirteen: the path reflectance, total transmittance and spherical albedo for each state and band, with the seven continuous state variables and the one-hot aerosol model and atmosphere profile as inputs [85]. Twelve of the thirteen bands keep a transmission above 0.14 on every state; band B10 at 1372 nm, in a strong water-vapour band, holds every smaller transmission. The splits use the same seeds and proportions (972 test, 7,875 fitting states).

The published libRadtran benchmark. The release of [65] pairs 6S and libRadtran values of the path reflectance a, the total transmittance t and the spherical albedo s for 50,000 states over the thirteen Sentinel-2 bands of [27], with its own state-level split into 35,000 training, 7,500 validation and 7,500 test states and its own code for the metrics. Its files hold 609,722 of the 650,000 rows of one state and one band, the rest removed by its quality control, and we fit and score on the rows present, 426,761, 91,490 and 91,471. Each row is fitted separately and the band enters through the inputs: eight state and band variables and the one-hot aerosol model and atmosphere profile, to which we add the 6S coefficients of the same row, and the network predicts the libRadtran coefficients minus the 6S ones, the residual form of the release's best models. The networks have five hidden layers of 512 SiLU units [30] and three outputs, are trained with Adam [47] and a one-cycle schedule [86] with a peak learning rate of $2 \cdot 10^{-3}$ for 300 epochs at batch size 2,048, and are kept at the epoch with the lowest validation value of their own objective. The weighted arms use the weights of J_u row by row at the true transmission, $\max\{t, \tau\}^{-2}$, $R^2 \max\{t, \tau\}^{-2}$ and $R^4 t^2 \max\{t, \tau\}^{-2}$ for the three coefficients with $\tau = uT_{\text{train}}$, each normalized to mean one on the training rows. Two variants locate the effect of the weights: the flattened weights with the spherical albedo left unweighted, and mixtures $\lambda + (1-\lambda)w$ of the flattened weights w with the plain objective, normalized in the same way. A kernel correction fits each network's residuals by Nyström kernel ridge regression [105] with the network's own row weights, on 4,096 landmarks with a Gaussian kernel on the standardized inputs, its bandwidth and ridge chosen on the validation value of the network's objective, and no correction where none improves it. Besides the release's split we use the out-of-distribution split of [85]: the test block holds the 13,847 states above the 0.85 quantile of the aerosol optical depth or of the water-vapour column, and the other 36,153 states are divided by the seed into 30,730 fitting and 5,423 validation states. The forward scores are the release's: the root-mean-square, mean absolute and symmetric mean absolute percentage errors over the three coefficients of every test row, and R^2 averaged over the coefficients. The retrieval scores are those below, with the top-of-atmosphere reflectance $a + \rho t/(1 - \rho s)$ in the role of the radiance.

Emulators. The members are the six families of [85]: a cubic ridge regression, Matérn kernel ridge regression on the inputs with an isotropic and with an input-scaled metric, a network of three hidden layers of 512 units, the network with a kernel correction of its residuals, and kernel ridge regression on the network's last hidden features. The kernel regressions are fitted on the full fitting block in the space of the leading 64 principal components of each standardized component, with their hyperparameters chosen on the validation block; the network is trained with AdamW and

a cosine schedule for up to 150 epochs and stopped on the validation loss with a patience of 25 epochs. The convex stacks of [106, 8] combine the six members with weights on a simplex for each component.

Scores. All scores are on the test block, at $\rho = 0.7$ unless stated otherwise: the mean relative ℓ^2 error of the radiance across bands, and the 95th percentile of the retrieval error $|\hat{\rho} - \rho|$, in points of reflectance (100 times the absolute error), over all entries and over the physical domain above fT_{train} for $f = 10^{-12}, 10^{-3}, 10^{-2}$, that is, the entries with $t \geq fT_{\text{train}}$, $0 \leq s < 1$ and $1 - \rho s \geq 0.3$. An inversion fails when the predicted denominator $\hat{t} + \hat{s}(L - \hat{a})$ is not positive or the retrieved value is not finite. Tables 1, 3, 4 and 6 follow [85]: a failed inversion is left out of the percentiles on the physical domain and its share is given beside them, and the all-band percentile takes a non-finite inversion as a retrieved reflectance of zero. Tables 2 and 5 score the principal networks, and on the EMIT table the emulators of [85], at $\rho = 0.1$, 0.4, 0.7 and 0.9 from their saved test predictions, and count every failed inversion as an infinite error, so that a tail is infinite once more than five percent of its entries fail.

5 Experiments

5.1 The two rankings on the EMIT table

The convex stack and the kernel on learned features of [85] are ordered differently by the two errors. On three splits at two network widths and on the physical domain above $10^{-3}T_{\text{train}}$, the first-order forward error at $\rho = 0.7$ favors the stack in all six runs, while the first-order retrieval error favors the kernel on features by factors of 5.0 to 9.9 and the exact constrained retrieval by factors of 3.3 to 4.5. The transmissions of the physical domain range over more than twenty orders of magnitude on this table, and the forward error, which weighs an entry by $g = t^2/q^4$ relative to the retrieval (Section 3), is blind to the entries that set the tail of the retrieval error. With $\Delta = d_f^2 - d_h^2$ for the stack f and the kernel h, all of the stack's net excess $\sum \Delta$ lies in the tenth of the entries with the smallest g, and the identity of Proposition 6.5, which splits the two orders, closes to rounding. The Pearson correlation of g and Δ is only -0.014, because Δ is heavy-tailed; the covariance, not the correlation, is the quantity the proposition needs.

5.2 Where the retrieval weight sits

A fit weighted for the retrieval learns, in effect, from about as many entries as lie below its floor, and most of its weight sits on entries that a retrieval at the same floor does not use (Proposition 6.8 and Corollary 6.9). On the EMIT table the share of the entries with transmission at most x behaves as Ax^β for small x, with $\beta \approx 0.12$, and the floored weights keep an effective fraction of 0.24, 0.11 and 0.07 of the entries at $u = 10^{-1}, 10^{-2}, 10^{-3}$, and 0.76, 0.90 and 0.92 of their weight lies below the floor, against 0.94, the population limit of Corollary 6.9 as $\tau \to 0$. The cut weights put none below the floor and keep an effective fraction of 0.13, 0.024 and 0.011.

The weight is concentrated in bands rather than in states. Write $\omega_i = \sum_b w_{ib}$ for the weight of state i and $\pi_{ib} = w_{ib}/\omega_i$ for its spread over the B bands. Over m states,

$$\frac{n_{\text{eff}}}{n} = \frac{(\sum_i \omega_i)^2}{m \sum_i \omega_i^2} \cdot \frac{\sum_i \omega_i^2}{B \sum_i \omega_i^2 \|\pi_i\|^2},$$

the effective fraction of the states times a weighted harmonic mean of the effective number of bands per state, $1/\|\pi_i\|^2$, divided by B; the identity is $n_{\text{eff}}/n = (\sum_{ib} w_{ib})^2/(mB \sum_{ib} w_{ib}^2)$ written out. On

the training block of one EMIT split the two factors are 0.73 and 0.095 at $u = 10^{-3}$, and 0.91 and 0.27 at $u = 10^{-1}$: the floored weights keep nearly three quarters of the states and about 27 of the 285 bands, which carry 90% of the weight. The eighteen bands 132–136 and 195–207, inside the water-vapour absorption near 1380 and 1870 nm, have more than half of their entries below $10^{-3}T_{\text{train}}$.

Where the weight sits is not where the value of the objective sits, because the absolute errors are smallest where the transmission is (Figure 1b). At the test residuals of the plain network of one EMIT split the entries below the floor carry 11% of the floored objective for the path radiance and 43% for the transmission at $u = 10^{-1}$, and 82% and 76% at $u = 10^{-3}$.

These are fractions of the weight against the physical squared error. The networks of Section 4 are fitted to outputs standardized band by band, and against their own loss the weight of an entry of the component c is $w_{ib}\sigma_{cb}^2$, with σ_{cb} the standard deviation of band b over the training states. The band variances differ by orders of magnitude. On the EMIT table they cut the entry effective fraction of the floored weights of the path radiance to 0.072, 0.021 and 0.017 at the three floors, a fifth to a quarter of the physical fraction below $u = 10^{-1}$, and the effective number of training states, $(\sum_i \omega_i)^2 / \sum_i \omega_i^2$, to 10,700, 7,200 and 8,900 of 18,883 (medians over ten splits; the direct and diffuse terms keep 5,700 to 10,100). The cut weights keep about 7,000 states of the path radiance at the two lower floors.

A floored weight gives its largest value, τ^{-2}, to every entry below the floor, and so to any evaluation of the forward code that failed and returned no transmission. In a band where the other transmissions are of the order of the median T, such an entry weighs as much as $(T/\tau)^2 = u^{-2}$ ordinary ones, a million at $u = 10^{-3}$, more than the table holds. The cut weights give it none, but an evaluation need not fail all at once. The two failed evaluations of the EMIT table (Section 4) fail by stages: in the five bands from 812 to 842 nm their transmission, a fifth down to a two-hundredth of the neighbours', is still above $10^{-2}T$, so the cut weights count these entries too, and their errors are as large as the fluxes themselves. Against the physical squared error they are two states in 23,313 and change none of the fractions above by more than one percent. In the coordinates of training, where the weight of an entry carries the variance of its band, they are the only states below the floor in more than 160 bands from 850 nm on, among them the near-infrared bands, where the path radiance varies more than anywhere outside the visible. On the eight splits that keep both in the training block they hold 16% of the floored weight of the path radiance at $u = 10^{-2}$ and 45% at $u = 10^{-3}$, and the effective number of training states falls to 75 and 10. Section 5.3 fits the networks with and without them.

5.3 Networks fitted on the retrieval error

The network of Section 4 is trained on each component with the weights of the first-order bound,

$$J_u = \mathbb{E}\Big[\frac{e_a^2 + R^2(e_{Y_2}^2 + e_{Y_3}^2) + R^4 t^2 e_s^2}{\max\{t, uT_{\text{train}}\}^2}\Big], \qquad R = 0.9,$$

normalized to mean one on the training block of each component and stopped on the same weighted loss of the validation block (the *floored* arms). Three variants change one thing each: the floored weights at $u = 10^{-2}$ stopped on the plain validation loss, the mixed weights $(1 - \alpha) + \alpha\bar{w}_\tau$ of Section 3 with $\alpha = 0.2$ at $u = 10^{-2}$, and the cut weights v_τ at $u = 10^{-2}$, normalized and used for stopping in the same way. The *flattened* arms raise the floored weights to a power $\kappa \in (0, 1)$, the device Shimodaira proposed for covariate shift [84], and normalize them to mean one in the same way: at $u = 10^{-2}$ with $\kappa = 1/4$ and $\kappa = 1/2$, and with $\kappa = 1/4$ stopped on the plain validation loss at $u = 10^{-2}$ and 10^{-3}. For the path radiance and the two fluxes the flattened weights at κ are, above

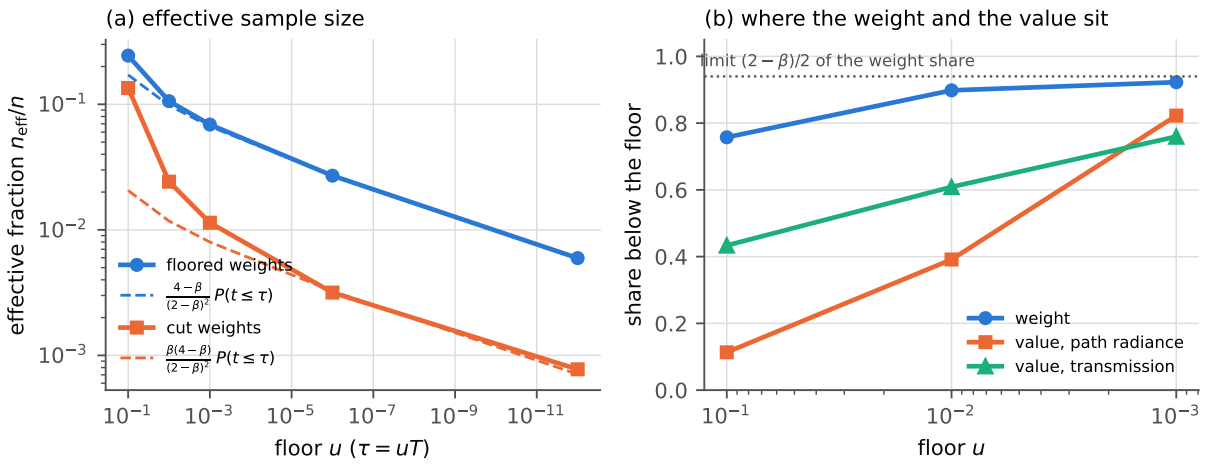


Figure 1: (a) The effective fraction n_{eff}/n of the floored weights $\max\{t,\tau\}^{-2}$ and of the cut weights $t^{-2}\mathbf{1}\{t>\tau\}$ over the entries of the physical domain of the EMIT table, $\tau = uT$ with T the median transmission, and the limits of Proposition 6.8 at $\beta = 0.12$. (b) On the test block of one EMIT split, the share of the floored weight on the entries below the floor, and the share of the floored objective's value there at the residuals of the plain network, for the path radiance and the transmission; the dotted line is the limit of Corollary 6.9.

the floor, those of the certificate of Proposition 6.6 at $\alpha = \kappa$; the albedo's become constant above the floor. We write K_u for the objective J_u with v_τ in place of the floored weights. The *joint* arms train one network for all four components on the first-order retrieval error at $\rho \in \{0.1, 0.4, 0.7\}$ with t floored at uT_{train}, plus 10^{-2} times the standardized squared error. Every run also fits the plain network on the same split with the same seed. Training is deterministic: on the table as provided the plain networks of the different runs agree to four decimals of the radiance error on every split, so every difference below is paired. Run on the table as provided, the code behind the upper block of Table 1 reproduces the runs of its lower block on splits 101 and 105, every arm to four decimals of every score on the same machine, so the two blocks differ in the two failed evaluations alone. Table 1 gives the differences on the EMIT table without its two failed evaluations and, in its lower block, on the table as provided. Without the failed evaluations the plain network has a radiance error of 0.350%, an all-band tail of 64.9 points and physical tails of 9.7, 3.5 and 2.1 points at the three floors.

Without the failed evaluations the floored weights at $u = 10^{-1}$ improve on the plain network in every tail on every split. The tails on the physical domain fall by nearly a quarter at $10^{-12}T_{\text{train}}$ and by more than a third at 10^{-3} and 10^{-2}, the all-band tail falls by 8.7 points, the share of failed inversions above $10^{-3}T_{\text{train}}$ falls from 0.10% to 0.04%, and the radiance error falls by 0.01 percentage points, on nine splits of ten. On the test blocks the network has the lower value of the floored objective at all three floors on every split, by median factors of 0.64, 0.42 and 0.56 at $u = 10^{-1}, 10^{-2}, 10^{-3}$, and of K_u at 10^{-2} and 10^{-3} on every split (0.36 and 0.42). The components move apart: the network lowers the error of the path radiance by a quarter and that of the direct flux by nearly a tenth, and it triples that of the albedo, from 1.6% to 4.8%. The albedo weights of J_u, $R^4t^2/\max\{t,\tau\}^2$, vanish with the transmission, and so does the fit: on split 101 the network's albedo error is 0.64% against 0.70% on the entries above $10^{-1}T_{\text{train}}$, and 11.5% against 3.8% on those between 10^{-2} and $10^{-1}T_{\text{train}}$.

arm	radiance [pp]	all bands	phys. 10^{-12}	phys. 10^{-3}	phys. 10^{-2}	failed [pp]
without the two failed evaluations						
floored, $u = 10^{-1}$	-0.01 ± 0.01 (9)	-8.7 ± 2.4 (10)	-2.3 ± 1.3 (10)	-1.3 ± 0.2 (10)	-0.7 ± 0.1 (10)	-0.07
floored, $u = 10^{-2}$	$+0.26 \pm 0.02$ (0)	-19.9 ± 2.6 (10)	-2.2 ± 1.2 (10)	-0.1 ± 0.2 (7)	$+0.6 \pm 0.1$ (0)	-0.10
floored, $u = 10^{-3}$	$+1.33 \pm 0.05$ (0)	$+3.1 \pm 1.9$ (0)	$+26.9 \pm 2.3$ (0)	$+15.2 \pm 1.0$ (0)	$+13.1 \pm 0.7$ (0)	-0.10
floored, $u = 10^{-2}$, plain stopping	$+0.26 \pm 0.01$ (0)	-19.9 ± 2.6 (10)	-2.2 ± 1.3 (10)	-0.1 ± 0.2 (6)	$+0.6 \pm 0.1$ (0)	-0.10
mixed, $\alpha = 0.2$, $u = 10^{-2}$	-0.00 ± 0.01 (5)	-6.5 ± 2.0 (10)	-1.4 ± 1.1 (9)	-0.4 ± 0.2 (10)	-0.1 ± 0.1 (8)	-0.08
flattened, $\kappa = 0.25$, $u = 10^{-2}$	-0.03 ± 0.01 (10)	-12.3 ± 2.6 (10)	-3.0 ± 1.4 (10)	-1.0 ± 0.2 (10)	-0.5 ± 0.1 (10)	-0.08
flattened, $\kappa = 0.5$, $u = 10^{-2}$	-0.03 ± 0.01 (10)	-24.8 ± 3.0 (10)	-4.3 ± 1.3 (10)	-1.5 ± 0.2 (10)	-0.7 ± 0.1 (10)	-0.10
flattened, $\kappa = 0.25$, $u = 10^{-2}$, plain stopping	-0.02 ± 0.01 (10)	-12.4 ± 3.0 (10)	-2.6 ± 1.7 (9)	-1.0 ± 0.2 (10)	-0.5 ± 0.1 (10)	-0.08
flattened, $\kappa = 0.25$, $u = 10^{-3}$, plain stopping	-0.02 ± 0.01 (10)	-23.4 ± 3.5 (10)	-3.0 ± 1.3 (10)	-1.0 ± 0.2 (10)	-0.4 ± 0.1 (10)	-0.09
cut, $u = 10^{-2}$	$+0.32 \pm 0.02$ (0)	$+5.6 \pm 1.8$ (0)	-6.5 ± 1.4 (10)	-0.8 ± 0.2 (10)	-0.0 ± 0.1 (6)	$+0.98$
joint, $u = 10^{-2}$	$+0.17 \pm 0.02$ (0)	-27.1 ± 1.8 (10)	-2.4 ± 1.4 (9)	-0.1 ± 0.2 (6)	$+0.5 \pm 0.1$ (0)	-0.10
joint, $u = 10^{-3}$	$+0.83 \pm 0.03$ (0)	-8.1 ± 2.9 (10)	$+23.6 \pm 2.0$ (0)	$+13.7 \pm 1.2$ (0)	$+11.8 \pm 0.9$ (0)	-0.10
the table as provided						
floored, $u = 10^{-1}$	$+0.55 \pm 0.16$ (0)	$+0.9 \pm 2.6$ (4)	$+2.2 \pm 1.6$ (2)	$+1.5 \pm 0.5$ (0)	$+1.3 \pm 0.4$ (0)	$+0.05$
floored, $u = 10^{-2}$	$+5.56 \pm 5.08$ (0)	$+18.6 \pm 45.5$ (1)	$+45.5 \pm 53.7$ (0)	$+38.9 \pm 59.5$ (0)	$+37.9 \pm 60.6$ (0)	$+0.02$
floored, $u = 10^{-3}$	$+18.55 \pm 1.77$ (0)	$+350.8 \pm 142.8$ (0)	$+397.8 \pm 137.0$ (0)	$+413.1 \pm 140.8$ (0)	$+403.6 \pm 144.8$ (0)	$+0.15$
joint, $u = 10^{-2}$	$+0.85 \pm 0.24$ (0)	-20.1 ± 5.3 (10)	$+2.7 \pm 4.5$ (3)	$+2.5 \pm 1.0$ (0)	$+2.7 \pm 0.8$ (0)	-0.09
joint, $u = 10^{-3}$	$+6.26 \pm 3.25$ (0)	$+28.0 \pm 56.1$ (0)	$+70.6 \pm 55.9$ (0)	$+67.4 \pm 59.8$ (0)	$+63.5 \pm 54.6$ (0)	-0.04

Table 1: Networks fitted on the retrieval error, EMIT: difference from the plain network of the same run in the radiance error and in the 95th percentiles of the retrieval error (points), mean ± standard deviation over splits, with the number of splits on which the arm is lower in parentheses, and the mean difference in the share of failed inversions on the physical domain above $10^{-3}T_{\text{train}}$. Floored: weights $\bar{w}_\tau$; mixed: $(1-\alpha)+\alpha\bar{w}_\tau$; flattened: $\bar{w}_\tau^{\kappa}$ renormalized; cut: v_τ; joint: one network for all four components. Upper block: the table without its two failed evaluations, splits 101–110; lower block: the table as provided, splits 101–110.

At the lower floors the objective and the tails part. The floored network at $u = 10^{-2}$ has the lower value of its objective on held-out states on every split (median factor 0.84) and trades one error for another: its all-band tail falls by 20 points and its tail above $10^{-12}T_{\text{train}}$ by 2.2, while its radiance error rises by 0.26 percentage points and its tail above $10^{-2}T_{\text{train}}$ by 0.55. At $u = 10^{-3}$ the objective is again lower on every split (0.54), and every error rises, the radiance error by 1.3 points and the physical tails by 13 to 27. The joint arms, one network for all four components, trade the same way and a little more cheaply: at $u = 10^{-2}$ the all-band tail falls by 27 points on every split for a rise of 0.17 percentage points in the radiance error, and at 10^{-3} the physical tails rise by 12 to 24 points. The objective is a mean, and at $u = 10^{-3}$ most of its value sits on the entries below the floor (Section 5.2); a network can lower it there while the 95th percentile, which the rest of the physical domain sets, grows. Nor are the lower floors attained best by the networks fitted on them. On held-out states the lowest $J_{10^{-2}}$ belongs to the network fitted at $u = 10^{-1}$ on nine splits of ten and to the mixed weights on the tenth, never to the network fitted on it (median factor 0.42 against 0.84), and the lowest $J_{10^{-3}}$ to a network fitted at $u = 10^{-2}$ on every split (0.15 against 0.54). The cut weights make the point most plainly: the network fitted on $K_{10^{-2}}$ has a higher value of it on held-out states than the plain network on nine splits of ten (median factor 1.21), against 0.36 for the floored network at $u = 10^{-1}$. In the coordinates of training the floored weights at $u = 10^{-1}$ keep an effective fraction of 0.046 to 0.072 of the entries of the first three components, against 0.016 to 0.024 at $u = 10^{-2}$ and 0.007 to 0.013 for the cut weights there, and at the size of this table the entries a higher floor keeps are worth more than a closer match between the objective fitted and the objective scored.

Stopping on the plain validation loss moves none of the mean differences at $u = 10^{-2}$ by more than 0.07 points (on a single split by at most 1.4 points, in the all-band tail), so the trade comes from the training weights and not from the stopping rule. The mixture with $\alpha = 0.2$ lowers the tails on average at an unchanged radiance error, by less than the floored weights at $u = 10^{-1}$ on

each of the five errors of the table. The cut weights appear to lower the tail above $10^{-12}T_{\text{train}}$ by 6.5 points, but a failed inversion is left out of the percentile. The cut network fails on 1.1% of the entries above $10^{-3}T_{\text{train}}$, against 0.1% for the plain network, and on more than 5% of those above $10^{-12}T_{\text{train}}$ on every split, so that its tail there is infinite once the failures are counted. Counting every failure as an infinite error, on split 101 the tail above $10^{-3}T_{\text{train}}$ is 3.26 points for the cut network, 3.23 for the plain one and 2.19 for the floored network at $u = 10^{-1}$, and 3.8%, 3.5% and 2.6% of the entries have an error above five points.

Flattening the weights does better than any floor. At $u = 10^{-2}$ and $\kappa = 1/2$, weights proportional to $1/\max\{t, \tau\}$ for the path radiance and the fluxes, the network lowers the radiance error on every split, from 0.350% to 0.322% on average (8%, split by split 5 to 10%), and every tail on every split: the all-band tail from 64.9 to 40.1 points and the physical tails from 9.7, 3.5 and 2.1 to 5.3, 2.0 and 1.4 points, by 44, 42 and 32% on average (split by split 28–57%, 35–44% and 26–37%). The share of failed inversions above $10^{-3}T_{\text{train}}$ falls from 0.10% to 0.005%. Against the floored network at $u = 10^{-1}$ it is lower in the radiance error, the all-band tail and the tails above 10^{-12} and $10^{-3}T_{\text{train}}$ on every split, and higher in the tail above $10^{-2}T_{\text{train}}$ on nine splits of ten (1.38 against 1.32 points on average). It lowers the errors of the path radiance, the direct flux and the diffuse flux on every split (medians 0.49% to 0.37%, 0.95% to 0.86% and 0.39% to 0.38%) and raises the albedo's on every split, from 1.56% to 2.24%, against a tripling at $u = 10^{-1}$. It also attains the weighted objectives on held-out states better than the networks fitted on them: of all the networks fitted on one component at a time it has the lowest value of $J_{10^{-2}}$, $K_{10^{-2}}$ and $K_{10^{-3}}$ on every split, of $J_{10^{-1}}$ on nine and of $J_{10^{-3}}$ on six, with median factors against the plain network of 0.19 for $J_{10^{-2}}$, where the network fitted on $J_{10^{-2}}$ reaches 0.84, and of 0.54 for $J_{10^{-1}}$, where the network fitted on it reaches 0.64. The weights with $\kappa = 1/4$ give the same radiance error and most of the gain in the tails (30, 30 and 25% on the physical domain), and stopping them on the plain validation loss moves each of these averages by at most five points. None of this is explained by the effective sample size alone. Over the entries of the table the flattened weights at $u = 10^{-2}$ keep an effective fraction of 0.37 at $\kappa = 1/4$ and 0.15 at $\kappa = 1/2$, against 0.11 for the floored weights at that floor and 0.24 at $u = 10^{-1}$: the best network keeps fewer effective entries than the floored network at $u = 10^{-1}$. What flattening keeps, and a higher floor gives up, is the order of the weights between 10^{-2} and $10^{-1}T_{\text{train}}$, where a floor at 10^{-1} makes them equal; what it removes is most of their range, a factor of 100 between 10^{-2} and 1 times T_{train} instead of 10^4.

The floor itself can be chosen on the validation block. A second set of runs trains the network on the floored weights at six floors, $u \in \{10^3, 1, 0.3, 0.1, 0.03, 0.01\}$, on each split, and keeps the one with the lowest cut objective $K_{10^{-2}}$ on the validation block, a rule fixed before the runs. On the EMIT table it picks $u = 10^{-1}$ on every split, the floor whose network also has the lowest tail above $10^{-3}T_{\text{train}}$ on the test block, and so it reproduces the first row of Table 1: the four tails fall by 8.7, 2.3, 1.3 and 0.73 points on every split at an unchanged radiance error. The floors on either side trade in opposite directions. At $u = 1$ and above the floored weights are nearly constant in the transmission and the fit is a least-squares fit in physical units: it lowers the radiance error by 0.10 percentage points on every split, from 0.35% to 0.25%, and raises the tails on average, the one above $10^{-12}T_{\text{train}}$ by 4 to 10 points. Weighting by the band variance alone does the same to within 0.05 points in every score, so the albedo weights play no part there. At $u = 0.03$ the all-band tail falls by 19 points and the radiance error rises by 0.08, the trade of $u = 10^{-2}$ on a smaller scale. The rule settles on the one floor that improves both.

The order of the networks does not depend on the reflectance at which the retrieval is scored, and it survives counting every failed inversion as an infinite error (Table 2). At $\rho = 0.1$, 0.4, 0.7 and 0.9 the floored network at $u = 10^{-1}$ lowers the all-band tail and the tails on the physical domain on every split, and the flattened network is lower again on every split, in the radiance error and in

	$\rho = 0.7$			tail, phys. 10^{-3}				failed [%], $\rho = 0.7$	
emulator	radiance [%]	all bands	phys. 10^{-12}	$\rho = 0.1$	0.4	0.7	0.9	phys. 10^{-12}	phys. 10^{-3}
networks of this paper									
plain	0.350	66.2	52.32	0.54	2.06	3.58	4.58	2.59	0.103
floored, $u = 10^{-1}$	0.342	58.7	39.13	0.34	1.27	2.19	2.81	2.31	0.038
flattened, $\kappa = 1/2$, $u = 10^{-2}$	0.322	41.1	21.40	0.31	1.17	2.04	2.61	2.08	0.005
emulators of [85]									
Matérn kernel, isotropic	0.232	69.9	69.26	1.32	5.11	8.87	11.37	3.21	0.600
Matérn kernel, input-scaled	0.075	38.0	22.98	0.28	0.88	1.48	1.89	2.00	0.230
network with kernel correction	0.120	48.1	26.99	0.34	1.28	2.22	2.84	2.09	0.049
kernel on features, width 512	0.076	7.8	4.85	0.13	0.48	0.82	1.05	1.37	0.000
kernel on features, width 2000	0.058	7.5	4.66	0.12	0.43	0.75	0.96	1.39	0.000
convex stack, width 512	0.069	30.1	15.16	0.21	0.71	1.22	1.55	1.90	0.011
convex stack, width 2000	0.056	18.4	9.54	0.17	0.55	0.93	1.19	1.73	0.001

Table 2: The EMIT table scored at four surface reflectances, from saved test predictions: the networks of Table 1 (without the two failed evaluations) and the emulators of [85] on the same ten splits and test blocks. The radiance error and the 95th percentiles of $|\hat{\rho} - \rho|$ in points over all bands and on the physical domain above $10^{-12}T_{\text{train}}$ at $\rho = 0.7$, the tail above $10^{-3}T_{\text{train}}$ at four reflectances, and the share of failed inversions on the physical domain at $\rho = 0.7$. Every failed inversion counts as an infinite error. Mean over splits 101–110.

each of these tails. Its tail above $10^{-3}T_{\text{train}}$ is 0.57 times the plain network's at each of the four reflectances (0.54 to 0.63 split by split), from 0.31 against 0.54 points at $\rho = 0.1$ to 2.61 against 4.58 at $\rho = 0.9$, and its tail above $10^{-12}T_{\text{train}}$ is 0.41 to 0.44 times the plain network's. Above $10^{-3}T_{\text{train}}$ the inversion fails on 0.10%, 0.04% and 0.005% of the entries for the plain, floored and flattened networks, and counting the failures moves no mean tail by more than 0.12 points. Deeper in the domain they set the tail. Between 2.1% and 2.6% of the entries above $10^{-12}T_{\text{train}}$ fail for the three networks, so that the 95th percentile there is the 97th to 97.5th percentile of the finite errors: at $\rho = 0.7$ it is 52.3, 39.1 and 21.4 points, against 9.7, 7.4 and 5.3 with the failures left out.

The emulators of [85], rescored from their saved predictions on the same splits and test blocks, keep the two rankings of Section 5.1 at every reflectance: the convex stack has the lower radiance error and the kernel on learned features the lower retrieval tails, on every split at both widths, except the radiance error at width 2000 and $\rho = 0.9$, which is lower for the stack on nine splits of ten. The kernel on features is the strongest emulator of the EMIT table for the retrieval. At width 2000 its tail above $10^{-3}T_{\text{train}}$ is 0.12 points at $\rho = 0.1$ and 0.96 at $\rho = 0.9$, below the flattened network's on every split at every reflectance; its all-band tail at $\rho = 0.7$ is 7.5 points against 41.1; and it has the fewest failed inversions, 1.4% of the entries above $10^{-12}T_{\text{train}}$ and almost none above $10^{-3}T_{\text{train}}$. The retrieval weights improve the network they train but do not close the distance to a kernel fitted on learned features, whose radiance error at width 2000 is 0.058% against 0.322%.

With the failed evaluations (lower block of Table 1) every floored and joint arm raises the radiance error and the tails above 10^{-3} and $10^{-2}T_{\text{train}}$ on every split, and only the joint arm at $u = 10^{-2}$ has a lower all-band tail on all ten. With the failed states the plain network has the lower value of the weighted objective itself on the test blocks of nine, nine and ten of the ten splits at $u = 10^{-1}, 10^{-2}, 10^{-3}$, by median factors of 4.3, 9.9 and 8.3, while on the training blocks of splits 101–103 the weighted network reaches 0.03 to 0.06 of the plain network's value: it fits the entries that carry its objective and does not generalize from them. Two effects of the failed states combine. At $u = 10^{-2}$ and 10^{-3} they leave the floored weights of the path radiance 75 and 10 effective training states (Section 5.2). At $u = 10^{-1}$ they hold under one percent of the weight, but

no smooth fit returns a zero flux between neighbours with ordinary ones, and their share of the value of the objective is far larger. On split 104, whose test block holds one of them, the ratio of the floored objective of the plain network's residuals to their plain mean square is 23 for the direct flux and 86 for the diffuse flux at $u = 10^{-1}$, against 0.12 to 0.42 and 0.24 to 0.55 on the other nine splits: that one state carries about 99% of the objective of each flux. The weighted network at $u = 10^{-1}$ loses its accuracy on those two components, with errors of 2.4% and 0.90% on split 101 against 0.86% and 0.40% without the failed states. The split that puts a failed state in the validation block, 105, is the worst of all, since the weighted stopping rule there is decided by that state; at $u = 10^{-2}$ its tail above $10^{-2}T_{\text{train}}$ rises by 208 points, against 7 to 44 on the other splits. The failed states also cost the plain network part of its accuracy. Removing them lowers its radiance error on each of the ten splits, by 0.035 percentage points on average, and its albedo error from 2.93% to 1.57%, while its physical tails move by less than a quarter of a point, up on some splits and down on others.

On the libRadtran table, which has no failed evaluations, the floored and joint arms raise the radiance error on every split, by 0.03 to 0.48 percentage points against a plain error of 0.356%, and the tail above $10^{-3}T_{\text{train}}$ on nine or ten splits of ten. There the floored weights put 86%, 99.8% and essentially all of their mass on band B10 at the three floors, so that the network is asked to fit one band of thirteen, the band where the retrieval is ill-conditioned, at the expense of the other twelve. The rule that chooses the floor on the validation block fails there. It picks $u = 10^{-2}$ on every split, the floor closest to its own criterion, and that network raises the radiance error by 0.06 percentage points and the tails above 10^{-3} and $10^{-2}T_{\text{train}}$ by 0.12 points on every split, and lowers only the all-band tail, by 2.6 points on nine splits. No floor improves on the plain network there in both the radiance error and the tails: at $u = 1$ and above the radiance error falls by 0.02 percentage points or less on average and the tail above $10^{-3}T_{\text{train}}$ moves by less than 0.11 points on every split, while the all-band tail rises by 34 points.

5.4 The published libRadtran benchmark

On the benchmark of [65] (Section 4) the plain network stands among the release's models in their own metrics (Table 3). Its RMSE over ten seeds, 0.00711, lies between that of sRTMnet [10] as trained in the release, 0.00691, and that of the release's third model, 0.00784; pKANrtm reaches 0.00619, and a residual network tuned for this split in [85] reaches 0.00581.

The floored and flattened weights trade one of the release's metrics against the others, on each of the ten seeds. The floored weights at $u = 10^{-1}$ and the flattened weights with $\kappa = 1/2$ at $u = 10^{-2}$ lower the mean absolute error to 0.00061 and 0.00051, 0.33 and 0.27 of pKANrtm's and below every model of the release, and the symmetric percentage error to 1.85% and 1.52%, against 3.91% for pKANrtm and 4.30% for the plain network. They raise the RMSE to 0.00772 and 0.00780, 1.25 and 1.26 times pKANrtm's and about that of the release's third model, and lower R^2, which averages over the three coefficients, to 0.990, above only the release's weakest baseline. The RMSE pools the three coefficients, and the spherical albedo sets it: the albedo's error rises from 0.0097 to 0.0133 and 0.0134, while the errors of the path reflectance and of the transmittance fall by factors of 3.3 and 3.6 and of 5.1 and 5.7. The albedo's retrieval weight vanishes with the transmission, as on the EMIT table (Section 5.3), and a single network with three outputs gives the capacity the other two weights ask for.

In the retrieval the weighted networks are better by a wide margin, again on every seed. The flattened network lowers the radiance error from 0.913% to 0.108%, the all-band 95th percentile from 4.63 to 0.35 points, the tails on the physical domain from 3.59, 3.15 and 3.06 to 0.32, 0.29 and 0.28 points, and the share of failed inversions above $10^{-3}T_{\text{train}}$ from 0.146% to zero on every

emulator	RMSE	MAE	R^2	SMAPE [%]	RMSE by coefficient		
					ρ_{path}	T	S
pKANrtm [65]	0.00619	0.00186	0.9947	3.91			
sRTMnet [10], trained in [65]	0.00691	0.00211	0.9942	4.28			
DA [65]	0.00784	0.00261	0.9935	4.95			
GPAC [65]	0.00865	0.00301	0.9928	5.58			
FC [65]	0.00948	0.00342	0.9919	6.13			
RF [65]	0.01075	0.00404	0.9906	7.02			
UMR [65]	0.01394	0.00546	0.9869	9.85			
6S as the emulator	0.09790	0.05348	0.6386	32.38	0.02065	0.16183	0.04623
network, plain	0.00711 ± 0.00033	0.00254 ± 0.00027	0.9942 ± 0.0003	4.30 ± 0.27	0.00111 ± 0.00008	0.00751 ± 0.00097	0.00967 ± 0.00030
network, floored, $u = 10^{-1}$	0.00772 ± 0.00012	0.00061 ± 0.00003	0.9901 ± 0.0003	1.85 ± 0.06	0.00033 ± 0.00002	0.00148 ± 0.00008	0.01329 ± 0.00021
network, flattened, $\kappa = 1/2$, $u = 10^{-2}$	0.00780 ± 0.00006	0.00051 ± 0.00003	0.9899 ± 0.0002	1.52 ± 0.06	0.00031 ± 0.00003	0.00133 ± 0.00011	0.01345 ± 0.00010
network, flattened, albedo unweighted	0.00772 ± 0.00043	0.00307 ± 0.00032	0.9939 ± 0.0005	4.68 ± 0.37	0.00148 ± 0.00020	0.00912 ± 0.00091	0.00964 ± 0.00039
network, 0.3 plain $+0.7$ flattened	0.00630 ± 0.00040	0.00162 ± 0.00058	0.9945 ± 0.0003	3.11 ± 0.74	0.00080 ± 0.00023	0.00468 ± 0.00182	0.00969 ± 0.00028

	radiance [%]	all bands	phys. 10^{-12}	phys. 10^{-3}	phys. 10^{-2}	failed [%]
6S as the emulator	20.565	75.55	75.55	74.04	73.81	0.000
network, plain	0.913 ± 0.108	4.63 ± 1.33	3.59 ± 0.76	3.15 ± 0.65	3.06 ± 0.62	0.146 ± 0.092
network, floored, $u = 10^{-1}$	0.132 ± 0.010	0.36 ± 0.02	0.33 ± 0.02	0.30 ± 0.02	0.29 ± 0.02	0.000 ± 0.000
network, flattened, $\kappa = 1/2$, $u = 10^{-2}$	0.108 ± 0.009	0.35 ± 0.03	0.32 ± 0.02	0.29 ± 0.02	0.28 ± 0.02	0.000 ± 0.000
network, flattened, albedo unweighted	1.143 ± 0.128	3.99 ± 0.42	3.50 ± 0.32	3.11 ± 0.27	3.03 ± 0.26	0.070 ± 0.063
network, 0.3 plain $+0.7$ flattened	0.550 ± 0.236	2.28 ± 0.99	1.96 ± 0.85	1.71 ± 0.72	1.66 ± 0.69	0.043 ± 0.053

Table 3: The published libRadtran benchmark of [65]: test scores on the release's split. Upper block, the release's metrics over the three coefficients of every test row, and the RMSE of each coefficient; the first seven rows are the values published in Table 3 of the release for its model and its six baselines. Lower block, the retrieval scores at $\rho = 0.7$: the mean relative error of the top-of-atmosphere reflectance over the bands of a state, the 95th percentiles of $|\hat{\rho} - \rho|$ in points over all rows and on the physical domain above fT_{train}, and the share of failed inversions above $10^{-3}T_{\text{train}}$. Mean $\pm$ standard deviation over ten seeds; 6S as the emulator is the release's low-fidelity code scored as it stands.

seed; the floored network at $u = 10^{-1}$ does nearly as well. 6S scored as the emulator has a radiance error of 20.6% and a tail of 76 points, so the networks learn a correction of it about two orders of magnitude smaller.

The two variants show where the gain comes from. With the spherical albedo left unweighted and the other two coefficients flattened, the network keeps the plain network's albedo and loses the rest: its radiance error, 1.14%, is above the plain network's on nine seeds of ten, and so are its errors in the path reflectance and the transmittance. The stopping rule takes the epoch with the lowest validation value of the summed objective, and with the albedo unweighted that epoch falls where the plain network's does, between 23 and 39 on nine seeds, long before the weighted path reflectance and transmittance are fitted; the flattened network is kept at epochs 289 to 297. A mixture that keeps three tenths of the plain objective improves on the plain network in the RMSE, the mean absolute and percentage errors, the radiance error and every tail, on every seed. Its RMSE, 0.00630, is within 2% of pKANrtm's, at a radiance error of 0.550% and an all-band tail of 2.28 points.

The kernel correction of the residuals helps where a network leaves structure in them. It lowers the plain network's RMSE from 0.00711 to 0.00684 and its radiance error by 0.09 percentage points, on every seed, and with the mixture it brings the RMSE to 0.00623, below pKANrtm's on four seeds of ten. For the floored and flattened networks the corrections it finds on their weighted validation objectives are too small to change any score in Table 3.

Outside the training range the weighted networks gain in every forward score but the albedo's (Table 4). On the split whose test states lie above the 0.85 quantile of the aerosol optical depth or of the water-vapour column, the flattened and floored weights lower the RMSE from 0.00860 to 0.00615 and 0.00605, the mean absolute error from 0.00381 to 0.00085 and 0.00101, the radiance error from 1.59% to 0.25% and 0.30%, and the retrieval tails by factors of 7.4 to 8.4, each on every

emulator	RMSE	MAE	R^2	SMAPE [%]	RMSE by coefficient		
					ρ_{path}	T	S
6S as the emulator	0.11448	0.06692	0.5221	36.67	0.02650	0.18835	0.05603
network, plain	0.00860 ± 0.00066	0.00381 ± 0.00024	0.9953 ± 0.0003	4.42 ± 0.71	0.00181 ± 0.00026	0.01234 ± 0.00153	0.00807 ± 0.00044
network, floored, $u = 10^{-1}$	0.00605 ± 0.00021	0.00101 ± 0.00010	0.9950 ± 0.0002	1.45 ± 0.04	0.00070 ± 0.00010	0.00330 ± 0.00077	0.00989 ± 0.00012
network, flattened, $\kappa = 1/2$, $u = 10^{-2}$	0.00615 ± 0.00010	0.00085 ± 0.00004	0.9947 ± 0.0002	1.22 ± 0.06	0.00070 ± 0.00006	0.00272 ± 0.00017	0.01028 ± 0.00018
network, flattened, albedo unweighted	0.00823 ± 0.00146	0.00373 ± 0.00124	0.9954 ± 0.0006	3.85 ± 1.02	0.00198 ± 0.00053	0.01140 ± 0.00350	0.00804 ± 0.00021
network, 0.3 plain + 0.7 flattened	0.00578 ± 0.00086	0.00176 ± 0.00086	0.9964 ± 0.0003	2.23 ± 0.84	0.00109 ± 0.00032	0.00555 ± 0.00255	0.00804 ± 0.00019

	radiance [%]	all bands	phys. 10^{-12}	phys. 10^{-3}	phys. 10^{-2}	failed [%]
6S as the emulator	25.570	106.43	106.43	106.58	106.52	0.000
network, plain	1.586 ± 0.207	7.67 ± 1.20	6.89 ± 1.15	6.18 ± 0.95	6.05 ± 0.92	0.135 ± 0.080
network, floored, $u = 10^{-1}$	0.303 ± 0.037	0.93 ± 0.13	0.88 ± 0.12	0.83 ± 0.11	0.82 ± 0.11	0.001 ± 0.001
network, flattened, $\kappa = 1/2$, $u = 10^{-2}$	0.248 ± 0.011	0.91 ± 0.05	0.87 ± 0.05	0.82 ± 0.04	0.80 ± 0.04	0.001 ± 0.001
network, flattened, albedo unweighted	1.484 ± 0.538	5.67 ± 2.16	5.23 ± 1.88	4.77 ± 1.69	4.69 ± 1.67	0.059 ± 0.079
network, 0.3 plain + 0.7 flattened	0.640 ± 0.383	2.94 ± 1.56	2.73 ± 1.52	2.51 ± 1.34	2.45 ± 1.31	0.016 ± 0.014

Table 4: The out-of-distribution split of the benchmark: the test block holds the 13,847 states above the 0.85 quantile of the aerosol optical depth or of the water-vapour column. The scores are those of Table 3. Mean ± standard deviation over five seeds, which change the fitting and validation states and the initialization; the test block is the same for all five.

one of five seeds. The albedo's error rises from 0.0081 to 0.0103 and 0.0099, and with it R^2 falls slightly. The release reports an RMSE of 0.01022 for pKANrtm on an out-of-distribution split of its own, which differs from this one, so the two are not compared.

The gains do not depend on the reflectance at which the retrieval is scored, and they survive counting every failed inversion as an infinite error (Table 5). At $\rho = 0.1$, 0.4, 0.7 and 0.9 the floored and the flattened networks lower the radiance error and the tails above 10^{-12} and $10^{-3}T_{\text{train}}$ on every one of the ten seeds of the release's split and of the five of the out-of-distribution split, and so does the mixture with three tenths of the plain objective. The tail of the flattened network above $10^{-3}T_{\text{train}}$ grows with the reflectance, from 0.11 points at $\rho = 0.1$ to 0.36 at $\rho = 0.9$ on the release's split, as does the plain network's, from 0.78 to 4.09, and it stays between 0.09 and 0.14 times the plain network's (0.13 to 0.16 times on the out-of-distribution split). The larger error of the weighted networks in the spherical albedo matters most at large reflectance, where $1 - \rho s$ is smallest, and at $\rho = 0.9$ the flattened network still has a radiance error of 0.112% against 0.941% for the plain one. Counting the failed inversions moves the mean tails above $10^{-3}T_{\text{train}}$ by at most 0.23 points, since inversions fail there on about 0.15% of the entries for the plain network and on almost none for the weighted ones. On the deeper domain above $10^{-12}T_{\text{train}}$ they fail on 0.91%, 0.55% and 0.54% of the entries of the release's split for the plain, floored and flattened networks, the weighted networks on fewer entries than the plain one on seven and eight seeds of ten, and with the failures counted the tails there are 4.63, 0.36 and 0.35 points at $\rho = 0.7$, against 3.59, 0.33 and 0.32 with the failures left out.

The exponent matters more than the floor (Table 6, three seeds each). At $\kappa = 1/2$ the floors $u = 10^{-3}$ and 10^{-1} give RMSEs of 0.00785 and 0.00780, against 0.00780 at 10^{-2}, and radiance errors of 0.162% and 0.098%. A flatter exponent keeps more of the albedo's weight on the states with small transmission. At $\kappa = 1/4$ the albedo's error, 0.0093, is slightly below the plain network's, the radiance error, 0.108%, is that of $\kappa = 1/2$, and the RMSE, 0.00500, 0.00520 and 0.00617 on the three seeds, is below pKANrtm's on each; the mean absolute and percentage errors are 3.8 and 2.7 times lower than pKANrtm's, and R^2 is higher than pKANrtm's on two seeds of three. Keeping a tenth of the plain objective gives RMSEs of 0.00579, 0.00591 and 0.00605, also below pKANrtm's on every seed, at a radiance error of 0.235%. Neither the exponent nor the mixture was chosen on validation data here, and three seeds do not settle a comparison this close. A wider and deeper network does not help: the plain network's RMSE rises to 0.00846, its validation objective is lowest at epochs 13 to 17, and the mixture reaches 0.00641. On the out-of-distribution split the same

		the release's split					out-of-distribution split				
			tail, phys.		failed [%], phys.			tail, phys.		failed [%], phys.	
ρ	network	radiance [%]	10^{-12}	10^{-3}	10^{-12}	10^{-3}	radiance [%]	10^{-12}	10^{-3}	10^{-12}	10^{-3}
0.1	plain	0.895	1.05	0.78	0.91	0.146	1.450	2.25	1.91	0.41	0.140
	floored, $u = 10^{-1}$	0.165	0.14	0.12	0.55	0.000	0.314	0.32	0.29	0.32	0.001
	flattened, $\kappa = 1/2$, $u = 10^{-2}$	0.140	0.13	0.11	0.54	0.000	0.286	0.33	0.30	0.30	0.001
	0.3 plain + 0.7 flattened	0.586	0.58	0.47	0.65	0.043	0.652	0.77	0.68	0.38	0.016
0.4	plain	0.864	2.81	1.97	0.91	0.146	1.488	4.90	4.07	0.40	0.137
	floored, $u = 10^{-1}$	0.124	0.22	0.18	0.55	0.000	0.276	0.57	0.52	0.32	0.001
	flattened, $\kappa = 1/2$, $u = 10^{-2}$	0.101	0.21	0.18	0.54	0.000	0.228	0.57	0.51	0.30	0.001
	0.3 plain + 0.7 flattened	0.522	1.39	1.06	0.65	0.043	0.597	1.79	1.53	0.38	0.016
0.7	plain	0.913	4.63	3.25	0.91	0.146	1.586	7.68	6.37	0.40	0.135
	floored, $u = 10^{-1}$	0.132	0.36	0.30	0.55	0.000	0.303	0.93	0.83	0.32	0.001
	flattened, $\kappa = 1/2$, $u = 10^{-2}$	0.108	0.35	0.29	0.54	0.000	0.248	0.91	0.82	0.30	0.001
	0.3 plain + 0.7 flattened	0.550	2.28	1.73	0.65	0.043	0.640	2.94	2.52	0.37	0.016
0.9	plain	0.941	5.82	4.09	0.90	0.145	1.642	9.46	7.87	0.40	0.134
	floored, $u = 10^{-1}$	0.136	0.44	0.37	0.54	0.000	0.316	1.15	1.04	0.32	0.001
	flattened, $\kappa = 1/2$, $u = 10^{-2}$	0.112	0.43	0.36	0.53	0.000	0.259	1.13	1.01	0.30	0.001
	0.3 plain + 0.7 flattened	0.566	2.86	2.17	0.65	0.043	0.666	3.69	3.16	0.37	0.015

Table 5: The benchmark scored at four surface reflectances, from the saved test predictions of the networks of Tables 3 and 4: the radiance error, the 95th percentiles of $|\hat{\rho} - \rho|$ in points on the physical domain above $10^{-12}T_{\text{train}}$ and $10^{-3}T_{\text{train}}$, counting every failed inversion as an infinite error, and the share of failed inversions there. Mean over the ten seeds of the release's split and the five of the out-of-distribution split.

network, run on one seed, behaves otherwise. Fitted without weights it is kept at epoch 299 of 400 and reaches an RMSE of 0.00555, below the weighted networks of Table 4, at a radiance error of 0.49% and an all-band tail of 2.19 points, still well above theirs; its mixture reaches 0.00551.

The weighted networks were still improving when training stopped: their lowest validation objective came at epochs 288 to 299 of 300 on every seed and on both splits, against epochs 20 to 41 for the plain network.

The result differs from that on the libRadtran table of Section 4, where no floor improved on the plain network in both the radiance error and the tails. The two settings differ in three ways at once: the benchmark trains on four and a half times as many states, its networks see the 6S coefficients and fit their residual, and they see one band of a state at a time, with the band among the inputs, where the networks of the table predict all bands of a state at once.

5.5 Kernel regression fitted on the retrieval error

For kernel ridge regression the floored retrieval loss of Section 6.3 gives a reweighted estimator with truncated weights, of the form studied in [61]: for each component and band,

$$\min_f \sum_i w_i\big(f(x_i) - y_i\big)^2 + \lambda\|f\|^2, \qquad \alpha_b = (K + \lambda W_b^{-1})^{-1} y_b,$$

with the entry weights of J_u (floored) or K_u (cut) at the true transmission. The weights change from band to band, so every band is its own solve, and to keep 285 solves per component affordable every arm is fitted on the same 6,000 training rows, the subsample on which the kernel's parameters are chosen, with the isotropic Matérn kernel and the smoothness, length scale and nugget the forward criterion chooses on the validation block. The weights are scaled to mean one in each band, so

network	RMSE	MAE	R^2	SMAPE [%]	RMSE of S	radiance [%]	all bands
flattened, $\kappa = 1/4$, $u = 10^{-2}$	0.00546 ± 0.00062	0.00049 ± 0.00002	0.9951 ± 0.0012	1.46 ± 0.04	0.00935 ± 0.00109	0.108 ± 0.006	0.38 ± 0.02
flattened, $\kappa = 3/4$, $u = 10^{-2}$	0.00764 ± 0.00023	0.00064 ± 0.00002	0.9903 ± 0.0006	1.75 ± 0.03	0.01312 ± 0.00040	0.159 ± 0.007	0.43 ± 0.02
flattened, $\kappa = 1/2$, $u = 10^{-3}$	0.00785 ± 0.00002	0.00063 ± 0.00003	0.9898 ± 0.0001	1.62 ± 0.08	0.01348 ± 0.00003	0.162 ± 0.009	0.51 ± 0.02
flattened, $\kappa = 1/2$, $u = 10^{-1}$	0.00780 ± 0.00004	0.00049 ± 0.00005	0.9899 ± 0.0001	1.60 ± 0.07	0.01345 ± 0.00007	0.098 ± 0.013	0.32 ± 0.04
albedo unweighted, $\kappa = 1/4$	0.00703 ± 0.00027	0.00247 ± 0.00020	0.9942 ± 0.0003	4.32 ± 0.29	0.00971 ± 0.00029	0.885 ± 0.121	4.26 ± 1.32
albedo unweighted, $\kappa = 3/4$	0.00968 ± 0.00053	0.00440 ± 0.00020	0.9927 ± 0.0010	5.48 ± 0.14	0.00978 ± 0.00058	1.646 ± 0.093	5.85 ± 0.77
0.1 plain $+$ 0.9 flattened	0.00592 ± 0.00013	0.00083 ± 0.00012	0.9944 ± 0.0002	2.03 ± 0.17	0.00999 ± 0.00015	0.235 ± 0.053	0.89 ± 0.05
0.5 plain $+$ 0.5 flattened	0.00691 ± 0.00038	0.00241 ± 0.00029	0.9942 ± 0.0004	3.97 ± 0.32	0.00972 ± 0.00030	0.861 ± 0.114	3.43 ± 0.80
wider network, plain	0.00846 ± 0.00058	0.00336 ± 0.00043	0.9930 ± 0.0001	5.31 ± 0.65	0.01037 ± 0.00017	1.241 ± 0.191	5.31 ± 0.93
wider network, albedo unweighted	0.00832 ± 0.00117	0.00337 ± 0.00074	0.9930 ± 0.0008	5.23 ± 0.16	0.01029 ± 0.00043	1.199 ± 0.419	3.93 ± 1.31
wider network, 0.3 plain $+$ 0.7 flattened	0.00641 ± 0.00070	0.00162 ± 0.00095	0.9943 ± 0.0003	2.93 ± 0.96	0.00986 ± 0.00021	0.544 ± 0.381	2.13 ± 1.67

Table 6: Ablations on the release's split, three seeds each: the exponent κ and the floor u of the flattened weights, the albedo left unweighted, the share of the plain objective in the mixture, and, in the last block, a wider network of six hidden layers of 1,024 units trained for 400 epochs. The scores are those of Table 3; the sixth column is the RMSE of the spherical albedo S. Mean $\pm$ standard deviation over the seeds.

that the nugget means the same in every arm, and the plain arm is the same solve with $w = 1$; the two forms of the normal equations, $(K + \lambda W_b^{-1})\alpha_b = y_b$ and $(W_b K + \lambda I)\alpha_b = W_b y_b$, give the same objectives to 10^{-13} on 300 rows. On the table without its two failed evaluations, over ten splits, the plain fit has a radiance error of 0.28% and tails of 24.8, 8.6 and 5.0 points on the physical domain (medians), against 0.075% and 1.39 points above $10^{-3}T_{\text{train}}$ for the input-scaled kernel of Section 5.7, which has its own length scales and is fitted on every training row.

The floored weights at $u = 10^{-1}$ lower every physical tail on every split, by 15, 13 and 9% at the median of the paired ratios, lower their own objective on held-out states on every split (median factor 0.75) and the share of failed inversions above $10^{-3}T_{\text{train}}$ from 0.67% to 0.42%, and raise the radiance error on every split, by 12%. At $u = 10^{-2}$ the radiance error rises by 27% and the tail above $10^{-12}T_{\text{train}}$ falls by 28%, while the one above $10^{-2}T_{\text{train}}$ rises by 4%, each on every split, and the objective falls to 0.28 of the plain fit's. At $u = 10^{-3}$ the objective falls to 0.08 of the plain fit's, and the radiance error rises by 44% and the tails above 10^{-3} and $10^{-2}T_{\text{train}}$ by 28 and 53%, again on every split. The cut weights at 10^{-2} lower the tails above 10^{-12} and $10^{-3}T_{\text{train}}$ by 61 and 23% and multiply the radiance error by 14. The kernel repeats what the networks of Section 5.3 show: its objective on held-out states falls at every floor, and the tails follow it only at the floor that keeps the most effective entries.

5.6 Stacks chosen on the retrieval error

The test states of each run are split in two halves; the weights of a stack are fitted on one half and scored on the other, both ways, so that no state enters both the fit and its evaluation and no member saw a test state. The *component stack* has one simplex per component fitted on the mean relative error of that component, the objective of the stack of [85]; the *retrieval stack* fits the four simplices jointly on the first-order retrieval error $\sum (v_\rho^\top e)^2$ at $\rho = 0.7$ over the entries of the physical domain with $t \geq 10^{-3}T_{\text{train}}$.

On the EMIT table, over twenty runs (ten splits at network widths 512 and 2000) and 40 held-out halves, the component stack reproduces the reversal: its radiance error is lower than that of the kernel on features on all 40 halves and its tail on the physical domain above $10^{-3}T_{\text{train}}$ larger on all 40. The retrieval stack has a tail below the component stack's on 39 of the 40 halves and a higher radiance error on all 40. Along the family of Proposition 6.12 the tail stays nearly flat while the radiance error falls until the weight on the forward error is close to one (Figure 2).

With the mixture chosen on the fitting half alone, as the lowest radiance error among the

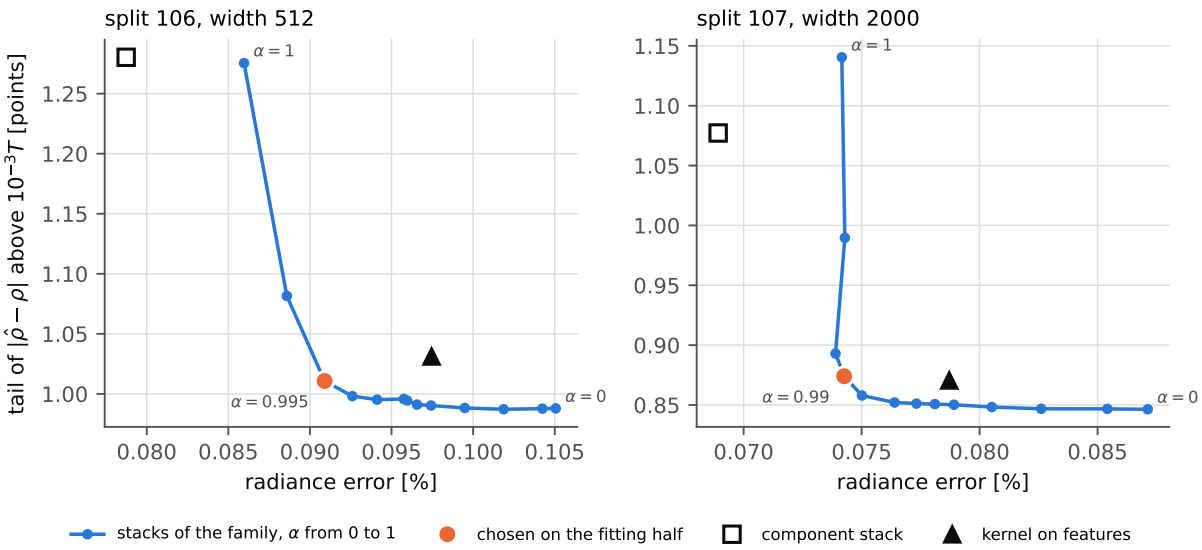


Figure 2: Stacks of the family of Proposition 6.12 on one held-out half of two EMIT runs, from the first-order retrieval error ($\alpha = 0$) to the forward error at $\rho = 0.7$ ($\alpha = 1$): the radiance error against the 95th percentile of the retrieval error on the physical domain above $10^{-3}T_{\text{train}}$. The orange point is the stack whose α is chosen on the other half; the square is the component stack and the triangle the kernel on learned features.

objectives whose tail on that half is no larger than that of the best single member, the chosen weight on the forward error is at least 0.9 on 36 of the 40 halves (median 0.99), and the stack has a lower radiance error than the kernel on features on 33 of the 40 held-out halves and a lower tail than the component stack on 39, by 25% at the median (quartiles 20% and 29%, at most 39%). Its tail lies between 0.06 points below and 0.22 points above that of the kernel on features. Refitted on the floored criterion instead, on every entry with $0 \le s < 1$ and with t floored at $10^{-3}T_{\text{train}}$, the stacks of splits 106–110 change their held-out tail by a factor between 0.995 and 1.014.

On the libRadtran table there is nothing to remove. Over the ten splits the component stack has a lower radiance error than the kernel on features on all 20 held-out halves and a smaller tail on 19, and the stack chosen by the same rule changes its tail by a factor between 0.93 and 1.05 and keeps it below the feature kernel's on all 20. Outside band B10 the density $g = t^2/q^4$ of Proposition 6.5 varies over a factor of 42 on this table, so the change of measure has little to reorder.

5.7 Kernel parameters chosen on the retrieval error

The kernel regressions of Section 4 choose their parameters on the validation block: the smoothness $\nu \in \{1.5, 2.5\}$, the length scale among four multiples of the median distance and the nugget among four values, and for the input-scaled kernel the relative length scales of the six inputs, by a coordinate search. The same grid and search can be run on a criterion for the retrieval instead of the forward error. The criteria for such choices are computed per component on the validation block, as the relative root-mean-square error weighted as in J_u: with the weights floored at uT_{train} (the *floored* criterion) or cut there, zero below (the *cut* criterion, with the weights v_τ of Section 3). The kernel is fitted with the chosen parameters on the plain squared error, and every run also chooses on the forward error, so that the differences are paired.

For the input-scaled kernel the retrieval criteria choose different parameters, and the choice pays

in the tail. On the EMIT table without its two failed evaluations, the cut criterion at $u = 10^{-2}$ lowers the tail above $10^{-3}T_{\text{train}}$ from 1.39 to 0.72 points and the all-band tail from 33 to 5.8 points, on each of the ten splits, and nearly removes the failed inversions above $10^{-3}T_{\text{train}}$ (0.23% to 0.002%), while the radiance error rises from 0.075% to 0.34%. The floored criterion at 10^{-2} and the cut one at 10^{-3} do about as well. The chosen metric moves toward the water-vapour column: its share of the input weight rises from 0.27 to 0.70 for the direct flux, from 0.26 to 0.54 for the diffuse flux and from 0.14 to 0.42 for the path radiance (medians over the ten splits). On the table as provided, split 105, whose validation block holds one of the failed evaluations, is where every choice goes wrong, the cut ones included, since that state's collapsing fluxes near 830 nm lie above the floor (Section 5.2); the floored criterion there picks a metric that raises the tail above $10^{-3}T_{\text{train}}$ by 18 points. For the isotropic kernel, which has no metric to move, every retrieval criterion raises the tail above $10^{-3}T_{\text{train}}$ on average, by 0.03 to 0.04 points at $u = 10^{-1}$ and by up to 6 points at 10^{-3}.

Choosing among a grid of kernels costs little in effective sample size, as Proposition 6.11 says. The input-scaled kernel chosen on the cut criterion at $u = 10^{-2}$ ends at about the radiance error of the networks of Section 5.3, 0.34% against 0.35% for the plain network and 0.34% for the network floored at $u = 10^{-1}$, lower than each on five splits of ten. Its tail above $10^{-3}T_{\text{train}}$ is 0.72 points against 3.5 and 2.2, lower on every split, about a fifth of the plain network's and a third of the floored network's (median ratios 0.20 and 0.33).

5.8 The stopping epoch

The stopping epoch of the plain network is a choice among the epochs of one training run, made here on the criteria of Section 5.7. On the EMIT table without its two failed evaluations, chosen by the cut criterion at $u = 10^{-1}$ instead of the plain validation loss, the stopping epoch lowers the radiance error on every split, by 0.005 percentage points on average, and the tail above $10^{-3}T_{\text{train}}$ on every split, by 0.06 points; the epoch has little to give. The criterion stops the two fluxes later, between epochs 141 and 150 against 106 to 150 for the plain rule, and at $u = 10^{-2}$ the cut and the floored criteria give the same small gains on every split. On the table as provided the choice at $u = 10^{-2}$ goes wrong on split 105, whose validation block holds one of the failed evaluations and whose entries both criteria count at that floor: the cut criterion stops the two fluxes at epochs 83 and 52 and the floored one at epochs 18 and 6, against 150 and 135 for the plain rule, and the radiance error rises by 0.26 and 1.24 percentage points.

6 Theory

This section gives the results behind the experiments of Section 5: the exact retrieval error and its metric (Section 6.1), the change of measure between the forward and retrieval rankings (Section 6.2), what a fit for the retrieval guarantees and what it costs (Section 6.3), and the price of choosing on the retrieval error (Section 6.4). The proofs are in Appendix A.

6.1 The retrieval error of an emulator

With the predicted components $(\hat{a}, \hat{t}, \hat{s}) = (a + e_a, t + e_t, s + e_s)$, the retrieval $\hat{\rho}$ and $q = 1 - \rho s$ of Section 3, the retrieval error has an exact closed form.

Lemma 6.1 (exact form of the retrieval error)**.** *Put*

$$\ell = -\frac{q}{t}\,(q\,e_a + \rho\,e_t) - \rho^2 e_s, \qquad \delta = \frac{q}{t}\,(e_t - s\,e_a) + \rho\,e_s - \frac{q}{t}\,e_a e_s.$$

Whenever the predicted denominator is nonzero,

$$\hat{\rho} - \rho = \frac{\ell + (\rho q/t)\, e_a e_s}{1 + \delta}.$$

Corollary 6.2 (first order)**.** *As the component errors tend to zero relative to t,*

$$\hat{\rho} - \rho = \ell + O\!\left(\frac{|e|^2}{t^2}\right), \qquad \ell = -\frac{q^2}{t}\big(\hat{L}(\rho) - L(\rho)\big) + O\!\left(\frac{|e|^2}{t^2}\right),$$

where $\hat{L}(\rho) = \hat{a} + \rho\hat{t}/(1 - \rho\hat{s})$ and $|e| = |e_a| + |e_t| + t|e_s|$. The retrieval error is the forward error at ρ divided by the slope $\partial_\rho L = t/q^2$.

Let ν be a probability measure on the reflectances of interest in $(0, R]$, and write $e = (e_a, e_t, e_s)^\top$ and

$$v_\rho = \left(\frac{q_\rho^2}{t},\ \frac{\rho q_\rho}{t},\ \rho^2\right)^\top, \qquad M(t,s) = \int v_\rho v_\rho^\top \, \nu(d\rho), \qquad q_\rho = 1 - \rho s.$$

By Corollary 6.2, $\int (\hat{\rho} - \rho)^2 \, \nu(d\rho) = e^\top M(t,s)\, e + O(|e|^3/t^3)$ at every entry, so the first-order mean squared retrieval error of an emulator over an evaluation distribution is the mean of a Mahalanobis norm of its component errors, with a metric that depends on the state and the band only through (t, s).

Proposition 6.3 (rank of the metric)**.** *If ν is a point mass, M has rank one. If the support of ν contains at least three points, M is positive definite for every $t > 0$ and $s \in [0, 1)$.*

Remark 6.4. At a single reflectance the retrieval sees one combination of the three component errors, $v_\rho^\top e$, and is blind to the two directions orthogonal to v_ρ. For small t the first two coordinates of v_ρ scale as $1/t$ and the third does not, so path-radiance and transmission errors are weighted by $1/t^2$ and the albedo error by a constant, in agreement with the transfer inequality $|\bar{\rho} - \rho| \le e_R/t$, $e_R = |e_a| + R|e_t| + R^2 t|e_s|$, of [85], where the albedo enters as $R^2 t|e_s|/t = R^2|e_s|$.

6.2 Forward and retrieval rankings differ by a change of measure

Fix ρ and write $d_f = v_\rho^\top e_f$ for an emulator f with component errors e_f. By Corollary 6.2, to first order the retrieval error is $\hat{\rho}_f - \rho = -d_f$ and the forward error is $r_f = \hat{L}_f(\rho) - L(\rho) = (t/q^2)\, d_f$. Let p be a distribution over the entries, a state and a band, with $\mathbb{E}_p[d_f^2] < \infty$, and put $g = t^2/q^4$.

Proposition 6.5 (change of measure)**.** *If $0 < \mathbb{E}_p[g] < \infty$, then $\mathbb{E}_p[r_f^2] = \mathbb{E}_p[g]\, \mathbb{E}_{p_g}[d_f^2]$ with $dp_g/dp = g/\mathbb{E}_p[g]$. Two emulators f, h are ordered by the forward squared error as they are ordered by the squared retrieval error under p_g. The orders under p and p_g disagree, $\mathbb{E}_p[r_f^2] < \mathbb{E}_p[r_h^2]$ while $\mathbb{E}_p[d_f^2] > \mathbb{E}_p[d_h^2]$, exactly when $\Delta = d_f^2 - d_h^2$ satisfies*

$$\mathbb{E}_p[\Delta] > 0 \quad \textit{and} \quad \mathrm{Cov}_p(g, \Delta) < -\mathbb{E}_p[g]\, \mathbb{E}_p[\Delta].$$

Relative to the retrieval, the forward metric gives an entry of transmission t a weight of order t^2, so an emulator whose excess squared error sits at small transmission can win on the forward metric and lose on the retrieval; Section 5.1 measures this on the EMIT table.

6.3 What a fit for the retrieval guarantees and what it costs

Consider one output coordinate $c \in \{a, t, s\}$ at one band as a scalar regression on the state x, with training distribution p on the physical domain $0 \le s < 1$, and write M^τ for the metric M of Section 6.1 with t replaced by $\max\{t, \tau\}$ for a floor $\tau > 0$. Since $0 < q_\rho \le 1$ and $\rho \le R$, its diagonal is bounded, $M^\tau_{aa} \le \tau^{-2}$, $M^\tau_{tt} \le R^2\tau^{-2}$ and $M_{ss} = \int \rho^4 \, \nu(d\rho) \le R^4$, so that $0 < \mathbb{E}_p[M^\tau_{cc}] < \infty$ and

$$\frac{dp^\tau_c}{dp}(x) = w^\tau_c(x) = \frac{M^\tau_{cc}(x)}{\mathbb{E}_p[M^\tau_{cc}]}$$

defines a probability distribution p^τ_c. The diagonal of the floored first-order retrieval risk is a multiple of the mean squared error under p^τ_c,

$$\mathbb{E}_p[M^\tau_{cc} e^2_c] = \mathbb{E}_p[M^\tau_{cc}] \, \mathbb{E}_{p^\tau_c}[e^2_c].$$

Fitting for the retrieval at a floor is therefore regression under covariate shift, with a likelihood ratio that is known in closed form rather than estimated and that is bounded, by $\tau^{-2}/\mathbb{E}_p[M^\tau_{aa}]$ for the path radiance and by $R^2\tau^{-2}/\mathbb{E}_p[M^\tau_{tt}]$ for the transmission; the off-diagonal terms of M^τ couple the coordinates but do not change this. For kernel ridge regression under a uniformly bounded ratio, the unweighted estimator with a suitable regularization is minimax optimal up to a logarithmic factor [61].

Without the floor the construction needs $\mathbb{E}_p[M_{cc}] < \infty$. On the physical domain $q_\rho \ge 1 - R > 0$, so that $M_{aa} \ge (1-R)^4 t^{-2}$ and $M_{tt} \ge (1-R)^2 \int \rho^2 \, \nu(d\rho) \, t^{-2}$, and for the path radiance and the transmission the condition is $\mathbb{E}_p[t^{-2}] < \infty$. Under the tail hypothesis of Proposition 6.8, $\mathbb{P}_p(t \le u) \sim Au^\beta$ with $0 < \beta < 2$, the proof of that proposition gives $\mathbb{E}_p[t^{-2}\mathbf{1}\{t > \tau\}] \to \infty$ as $\tau \to 0$, so $\mathbb{E}_p[t^{-2}] = \infty$: the unfloored retrieval weights of these two coordinates define no distribution, and as $\tau \to 0$ a fraction $(2-\beta)/2$ of the floored weight comes to lie below the floor (Corollary 6.9). The minimax rate that [61] prove for a reweighting truncated at a suitable level assumes an unbounded ratio with a finite second moment, which needs a finite normalization to begin with; it does not cover the unfloored retrieval weights. A retrieval loss with a floor τ is therefore not an approximation to a finite unfloored target but regression under the bounded shift p^τ_c itself, with a ratio the physics supplies.

On a finite table every one of these means is finite as long as every transmission is positive, and the effective sample sizes of Section 5.2 are quantities of the table. The divergence concerns the population: on a table it shows as weight concentrated on the smallest transmissions, and Proposition 6.8 and Corollary 6.9 give its limit as $\tau \to 0$ under the tail hypothesis, which the transmissions of the EMIT table follow with $\beta \approx 0.12$. An entry with $t = 0$, such as the two failed evaluations of Section 4, makes even the table's unfloored mean infinite.

6.3.1 The rate a weighted fit guarantees

Write $e_R = |e_a| + R|e_t| + R^2 t|e_s|$ and $\bar\rho$ for the retrieval constrained to $[0, R]$, which satisfies $|\bar\rho - \rho| \le \min\{R, e_R/t\}$ at every entry [85].

Proposition 6.6 (the rate a weighted certificate buys)**.** *Let $0 \le \alpha \le 1$ and $\mathbb{E}_p[t^{-2\alpha} e^2_R] \le \varepsilon^2$.*

1. *If $\alpha = 1$, then $\mathbb{E}_p|\bar\rho - \rho|^2 \le \varepsilon^2$.*
2. *If $\alpha < 1$ and $\mathbb{P}_p(t \le u) \le Au^\beta$ for $0 < u \le u_0$, then for $\varepsilon \le u_0^{(\beta+2-2\alpha)/2}$,*

$$\mathbb{E}_p|\bar\rho - \rho|^2 \le (R^2 A + 1) \, \varepsilon^{2\beta/(\beta+2-2\alpha)}.$$

Proposition 6.7 (the weighted exponent is attained)**.** *Let $\beta > 0$, $0 \le \alpha < 1$, $0 < c < r_0 < R$ and $\tau \in (0,1)$. Take $s = a = 0$, $\rho = r_0$, $\mathbb{P}_p(t \le u) = u^\beta$ on $(0,1]$, and the predictor $\hat{t} = t$, $\hat{s} = 0$, $\hat{a} = c\,t\,\mathbf{1}\{t \le \tau\}$. Then $\bar{\rho} - \rho = -c\,\mathbf{1}\{t \le \tau\}$,*

$$\mathbb{E}_p[t^{-2\alpha} e_R^2] = \frac{c^2 \beta}{\beta + 2 - 2\alpha}\, \tau^{\beta+2-2\alpha} =: \varepsilon^2,$$

$$\mathbb{E}_p|\bar{\rho} - \rho|^2 = c^2 \tau^\beta = c^{\frac{4-4\alpha}{\beta+2-2\alpha}} \Big(\frac{\beta + 2 - 2\alpha}{\beta}\Big)^{\frac{\beta}{\beta+2-2\alpha}} \varepsilon^{\frac{2\beta}{\beta+2-2\alpha}}.$$

At $\alpha = 1$ the same predictor gives $\mathbb{E}_p|\bar{\rho} - \rho|^2 = \mathbb{E}_p[t^{-2} e_R^2]$, so part 1 of Proposition 6.6 holds with equality.

At $\alpha = 0$ the exponent is the $2\beta/(\beta+2)$ of plain fitting, and it rises continuously to 2 as $\alpha \to 1$. If the plain fit reaches $\mathbb{E}_p[e_R^2]^{1/2} = O(n^{-r_0})$ and the fit certified at α reaches $\mathbb{E}_p[t^{-2\alpha} e_R^2]^{1/2} = O(n^{-r_\alpha})$, weighting gives the better guarantee exactly when $r_\alpha > r_0(\beta + 2 - 2\alpha)/(\beta + 2)$, a factor close to $1 - \alpha$ for small β; at the EMIT value $\beta \approx 0.12$ and $\alpha = 1/2$ it is 0.53.

6.3.2 The price of the weighting

For entry weights w the effective sample size of a weighted fit on n entries is $n\,\mathbb{E}_p[w]^2/\mathbb{E}_p[w^2]$ [48], and for the weights of the retrieval it is set by the transmission tail.

Proposition 6.8 (the effective sample size of retrieval weighting)**.** *Let $0 < \beta < 2$ and suppose $F(u) = \mathbb{P}_p(t \le u)$ satisfies $F(u)/(Au^\beta) \to 1$ as $u \to 0$, with $\mathbb{E}_p[t^{-4}\mathbf{1}\{t > u_0\}] < \infty$ for some $u_0 > 0$. For the floored weights $w_\tau = \max\{t, \tau\}^{-2}$ and the cut weights $v_\tau = t^{-2}\mathbf{1}\{t > \tau\}$, as $\tau \to 0$,*

$$\frac{\mathbb{E}_p[w_\tau]^2}{\mathbb{E}_p[w_\tau^2]} \sim \frac{4 - \beta}{(2 - \beta)^2}\, F(\tau), \qquad \frac{\mathbb{E}_p[v_\tau]^2}{\mathbb{E}_p[v_\tau^2]} \sim \frac{\beta(4 - \beta)}{(2 - \beta)^2}\, F(\tau).$$

Corollary 6.9 (where the floored weight sits)**.** *Under the hypotheses of Proposition 6.8,*

$$\frac{\mathbb{E}_p[w_\tau \mathbf{1}\{t \le \tau\}]}{\mathbb{E}_p[w_\tau]} \longrightarrow \frac{2 - \beta}{2} \qquad (\tau \to 0).$$

A fit weighted for the retrieval therefore learns, in effect, from about as many entries as lie below its floor, and most of its weight sits on entries that a retrieval at the same floor does not use; Section 5.2 measures both on the EMIT table.

Mixing the two objectives buys effective sample size back once α is small against $\sqrt{\rho_\tau}$. With $\bar{w}_\tau = w_\tau/\mathbb{E}_p[w_\tau]$ and $\rho_\tau = 1/\mathbb{E}_p[\bar{w}_\tau^2]$, the weights $w = (1 - \alpha) + \alpha \bar{w}_\tau$ have mean one and effective fraction $1/(1 - \alpha^2 + \alpha^2/\rho_\tau)$, while $\mathbb{E}_p[\bar{w}_\tau e_R^2] \le \mathbb{E}_p[w\, e_R^2]/\alpha$ keeps the retrieval certificate up to the factor $1/\alpha$. For the path radiance in the coordinates of training at $u = 10^{-2}$, where $\rho_\tau = 0.021$, the fraction is 0.89, 0.35 and 0.08 at $\alpha = 0.05$, 0.2 and 0.5.

Corollary 6.10 (rate against effective sample size)**.** *Under the hypotheses of Proposition 6.8, suppose a plain fit on n entries has $\mathbb{E}_p[e_R^2] \le C n^{-\gamma}$ and a fit with the normalized floored weights $\bar{w}_\tau$ has $\mathbb{E}_p[\bar{w}_\tau e_R^2] \le C n_{\text{eff}}^{-\gamma}$ with $n_{\text{eff}} = n\,\mathbb{E}_p[w_\tau]^2/\mathbb{E}_p[w_\tau^2]$, for some $\gamma > 0$. With the floor chosen as a power of n,*

$$\mathbb{E}_p|\bar{\rho} - \rho|^2 = O\big(n^{-\beta\gamma/(2+\beta)}\big) \text{ (plain)}, \qquad \mathbb{E}_p|\bar{\rho} - \rho|^2 = O\big(n^{-\beta\gamma/(2+\beta\gamma)}\big) \text{ (floored weights)},$$

so the weighted exponent is the larger exactly when $\gamma < 1$.

On EMIT the learning curves of [85] give mean-square exponents of about 0.8 to 1.2 on the first three components, and with $\beta \approx 0.12$ the two exponents agree to within about one percent either way. Rates cannot decide whether weighting pays on this table; the constants decide it, and Section 5.3 measures them.

6.4 Choosing on the retrieval error

Fitting a model on a weighted objective pays the effective sample size of the entries. Choosing among a few candidates fitted on the forward error pays much less, because the choice is made on a sum over states.

Proposition 6.11 (choosing on a weighted criterion)**.** *Let $Z_1, \dots, Z_m$ be independent copies of a state Z with B entries, let $v_b(Z) \ge 0$ be entry weights with state weight $\omega(Z) = \sum_b v_b(Z) \le \omega_{\max}$, and let $\ell_{k,b}(Z) \in [0, L]$ be the entry losses of K candidates. Put $R_k = \mathbb{E} \sum_b v_b \ell_{k,b}$, let $\hat{R}_k$ be its mean over the m states, and let $m_{\mathrm{eff}} = m\,\mathbb{E}[\omega]^2/\mathbb{E}[\omega^2]$. With probability at least $1-\delta$ the candidate $\hat{k}$ that minimizes $\hat{R}_k$ satisfies*

$$R_{\hat{k}} - \min_k R_k \le 2L\,\mathbb{E}[\omega]\Big(\sqrt{\frac{2\log(2K/\delta)}{m_{\mathrm{eff}}}} + \frac{2\omega_{\max}}{3\,\mathbb{E}[\omega]}\,\frac{\log(2K/\delta)}{m}\Big).$$

The weights enter only through the effective number of states and the ratio $\omega_{\max}/\mathbb{E}[\omega]$. On an EMIT validation block of $m = 2{,}098$ states the cut weights keep 1,962, 1,831 and 1,639 effective states at $u = 10^{-1}, 10^{-2}, 10^{-3}$, with $\omega_{\max}/\mathbb{E}[\omega]$ between 2.6 and 3.0, and the floored weights 1,917, 1,763 and 1,579. For $K = 32$ and $\delta = 0.05$ the bound is 0.18 to 0.20 times $L\,\mathbb{E}[\omega]$, against 0.17 with no weights at all: weighting the criterion raises the price of choosing by eight to eighteen percent, while it divides the effective data of a weighted fit by fourteen at $u = 10^{-3}$.

A convex stack is a choice of this kind with a continuum of candidates. For members with component errors $e^{(j)}$ and weights w on a product W of simplices, write $F(w)$ and $G(w)$ for the first-order forward and retrieval errors of the stack; both are convex quadratics in w.

Proposition 6.12 (the mixtures trace the frontier)**.** *Let W be a product of simplices and $F(w) = w^\top B w$, $G(w) = w^\top A w$ with A, B positive semidefinite. If $w^* \in W$ is weakly Pareto optimal for (F, G), that is, no $w \in W$ has $F(w) < F(w^*)$ and $G(w) < G(w^*)$, then w^* minimizes $\alpha F + (1-\alpha)G$ over W for some $\alpha \in [0, 1]$.*

Lemma 6.13 (the path is monotone)**.** *For $0 \le \alpha_1 < \alpha_2 \le 1$ let w_i minimize $\alpha_i F + (1-\alpha_i)G$ over W. Then $F(w_2) \le F(w_1)$ and $G(w_2) \ge G(w_1)$.*

The proof of Proposition 6.12 uses only that W is convex and that F and G are convex on it, so the same holds for any convex class of predictors, for instance the functions of norm at most r in a reproducing kernel Hilbert space; kernel ridge regression with the entry weights $(1-\alpha) + \alpha g/\bar{g}$ is the penalized form of that problem.

7 Discussion

The retrieval error of an emulator is the forward error seen through a change of measure that favors small transmissions, and a table with entries deep in absorption bands makes the two rankings disagree. Fitting on the retrieval error follows the change of measure as far as the floor lets it, and the floor sets the price. On the EMIT table a floor at a tenth of the median transmission keeps enough effective entries for the fit to generalize, and the network trained on it improves on the plain one in every tail on every split; lower floors still lower their objective on held-out states but give up radiance error and, at the lowest, the tail. Flattening is the better way to temper the weights. The square root of the weights floored at a hundredth keeps the order of the small transmissions that a floor at a tenth erases and removes most of their range, and the network trained on it improves on the plain one in the radiance error and in every tail, on every split; the albedo, whose retrieval

weight vanishes with the transmission below the floor, is the one component it fits less well. The order of the plain, floored and flattened networks holds at every reflectance from 0.1 to 0.9 with the failed inversions counted, and none of them reaches the kernel on learned features of [85], which has the lowest retrieval tails on the EMIT table at every reflectance; the retrieval weights have not been tried on such a kernel. Kernel ridge regression on the floored weights moves as the networks do, so what is learned here concerns the weights and not the model class. On the published benchmark the weights carry over: the retrieval gains hold on every seed, on states beyond the training range of aerosol and water vapour the weighted networks also have the lower root-mean-square error, and in an ablation over three seeds a flatter exponent kept the albedo's accuracy with the same retrieval gain. The floor is therefore a parameter to be chosen on held-out states, and the lowest floor a fit can afford is set by the effective sample size of its weights rather than by the floor at which the retrieval is scored. A floored weight also magnifies whatever lies below the floor. Two failed evaluations of the code, two states in twenty-three thousand, turn the network at that floor from better than the plain one in every tail on every split to worse in the radiance error and the tail on every split, and a table meant for weighted training should first be screened for entries with zero transmission. Choosing on the retrieval error keeps the forward fit and moves only what the change of measure needs to move. For the stacks that removes most of the reversal, and choosing the mixture near the forward end of the family keeps the radiance error below that of the best single member on 33 of the 40 halves.

The first-order bound behind every criterion here is accurate where the component errors are small relative to the transmission, and the floors keep the criteria finite where they are not; the constrained retrieval and the transfer inequality of [85] cover the rest. The criteria are fitted at $\rho = 0.7$ or over three reflectances, and the fits keep their order when the retrieval is scored at 0.1, 0.4 and 0.9; a retrieval over a different range of reflectances changes the metric but not the argument.

Acknowledgements

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), by the Israel Science Foundation (ISF Grant 4101/25), and by Google Cloud through its education credits program (credit 530550103).

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [108] and working together inside the Brain framework [110], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [109], the Math Brain repository [108] and the Brain framework repository [110].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital; both entities supported this work and are named above. These entities and the public funders had no role in the study design, analysis, manuscript preparation, or decision to publish.

AI-assisted contributions

Claude Code (Anthropic) derived and checked the propositions, implemented and ran the experiments, produced the tables and figures from the run records, and drafted the manuscript. The author supplied the data, set the questions, and reviewed the results, the proofs and the text.

Data and code availability

The repository `https://github.com/yspennstate/rt-emulators-for-retrieval` holds the code, one JSON record per run with the digests of its inputs, the scripts that produce every table and figure from those records, and this manuscript. The EMIT table is the lookup table of [85], supplied by the Jet Propulsion Laboratory and available from the author on request, subject to the permission of the Jet Propulsion Laboratory. The libRadtran table and the benchmark of Section 5.4 are built from the public release accompanying [65].

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [108].

A Proofs

A.1 The retrieval error of an emulator

Proof of Lemma 6.1. Let $u = L - a = \rho t/q$ and $D = t + su = t/q$, so that $\rho = u/D$. The predicted denominator is $\hat{D} = \hat{t} + \hat{s}(u - e_a) = D + e_t + u\,e_s - s\,e_a - e_a e_s$, and

$$\hat{\rho} - \rho = \frac{u - e_a}{\hat{D}} - \frac{u}{D} = \frac{-e_a D - u(e_t + u\,e_s - s\,e_a) + u\,e_a e_s}{D\hat{D}}.$$

Since $D - su = t$, the numerator equals $-t\,e_a - u\,e_t - u^2 e_s + u\,e_a e_s$. Divide numerator and denominator by $D^2 = t^2/q^2$ and use $u/D = \rho$, $uq^2/t^2 = \rho q/t$ and $\hat{D}/D = 1 + \delta$. □

Proof of Corollary 6.2. The first statement is Lemma 6.1 with $|\delta| = O(|e|/t)$. For the second, the forward error is $\hat{L}(\rho) - L(\rho) = e_a + \rho e_t/q + \rho^2 t e_s/q^2 + O(|e|^2)$, and multiplying by $-q^2/t$ gives ℓ. □

Proof of Proposition 6.3. A point mass gives $M = v_\rho v_\rho^\top$. In general $x^\top M x = \int (v_\rho^\top x)^2\,\nu(d\rho)$, and $v_\rho^\top x = p_x(\rho)$ with $p_x(\rho) = x_1(1 - 2s\rho + s^2\rho^2)/t + x_2(\rho - s\rho^2)/t + x_3\rho^2$, a polynomial of degree at most two. Its coefficients in the basis $(1, \rho, \rho^2)$ are $(x_1/t,\ (x_2 - 2sx_1)/t,\ (s^2 x_1 - s x_2)/t + x_3)$, a triangular invertible map of x, so $p_x \equiv 0$ only for $x = 0$. A nonzero polynomial of degree at most two vanishes at no more than two points, hence $x^\top M x > 0$ for $x \neq 0$ once ν charges three. □

A comparison with the exact inverse on 10^5 random states, with component errors of the order of the components themselves, reproduces Lemma 6.1 to within 1.5×10^{-12}.

A.2 The change of measure

Proof of Proposition 6.5. The first identity is $r_f^2 = g\,d_f^2$ integrated against p. The second statement follows from $\mathbb{E}_p[g\Delta] = \mathbb{E}_p[g]\mathbb{E}_p[\Delta] + \mathrm{Cov}_p(g, \Delta)$. □

A.3 What a fit for the retrieval guarantees and what it costs

Proof of Proposition 6.6. For every $\tau > 0$ the pointwise bound gives $\mathbb{E}_p|\bar{\rho} - \rho|^2 \le R^2\mathbb{P}_p(t \le \tau) + \mathbb{E}_p[t^{-2}e_R^2\mathbf{1}\{t > \tau\}] \le R^2\mathbb{P}_p(t \le \tau) + \tau^{2\alpha-2}\varepsilon^2$, since $t^{-2} \le t^{-2\alpha}\tau^{2\alpha-2}$ on $\{t > \tau\}$. At $\alpha = 1$ let $\tau \to 0$ and use $t > 0$. Otherwise take $\tau = \varepsilon^{2/(\beta+2-2\alpha)} \le u_0$; then τ^β and $\tau^{2\alpha-2}\varepsilon^2$ both equal $\varepsilon^{2\beta/(\beta+2-2\alpha)}$. □

Proof of Proposition 6.7. With $s = 0$ the radiance is $L = r_0 t$, and the retrieval gives $(L - \hat{a})/\hat{t} = r_0 - c\,\mathbf{1}\{t \le \tau\}$, which lies in $[0, R]$. The only error is e_a, hence $e_R = c\,t\,\mathbf{1}\{t \le \tau\}$ and $\mathbb{E}_p[t^{-2\alpha}e_R^2] = c^2\int_0^\tau u^{2-2\alpha}\beta u^{\beta-1}\,du$. Solving for τ and substituting into $c^2\tau^\beta$ gives the second identity. □

Proof of Proposition 6.8. Integrating by parts, $\mathbb{E}_p[t^{-k}\mathbf{1}\{t > \tau\}] = -\tau^{-k}F(\tau) + k\int_\tau^\infty u^{-k-1}F(u)\,du$ for $k = 2, 4$. As $\tau \to 0$ the integral over $[u_0, \infty)$ stays bounded, while over (τ, u_0) it is $\frac{A}{k-\beta}\tau^{\beta-k}(1 + o(1))$ times k, which diverges because $\beta < 2 \le k$. Hence $\mathbb{E}_p[v_\tau] \sim A\frac{\beta}{2-\beta}\tau^{\beta-2}$ and $\mathbb{E}_p[v_\tau^2] \sim A\frac{\beta}{4-\beta}\tau^{\beta-4}$, and adding $\tau^{-2}F(\tau)$ and $\tau^{-4}F(\tau)$ gives $\mathbb{E}_p[w_\tau] \sim A\frac{2}{2-\beta}\tau^{\beta-2}$ and $\mathbb{E}_p[w_\tau^2] \sim A\frac{4}{4-\beta}\tau^{\beta-4}$. □

Proof of Corollary 6.9. The numerator is $\tau^{-2}F(\tau) \sim A\tau^{\beta-2}$ and the denominator is $A\frac{2}{2-\beta}\tau^{\beta-2}(1 + o(1))$ by the proof of Proposition 6.8. □

Proof of Corollary 6.10. For both fits $\mathbb{E}_p|\bar{\rho} - \rho|^2 \le R^2F(\tau) + \mathbb{E}_p[t^{-2}e_R^2\mathbf{1}\{t > \tau\}]$. For the plain fit the second term is at most $\tau^{-2}Cn^{-\gamma}$, and $\tau = n^{-\gamma/(2+\beta)}$ balances it against $R^2F(\tau) \sim R^2A\tau^\beta$. For the weighted fit $t^{-2}\mathbf{1}\{t > \tau\} \le w_\tau$, so the second term is at most $\mathbb{E}_p[w_\tau]\,Cn_{\text{eff}}^{-\gamma}$, which by Proposition 6.8 is of order $\tau^{\beta-2}(n\tau^\beta)^{-\gamma}$; $\tau = n^{-\gamma/(2+\beta\gamma)}$ balances it. □

A.4 Choosing on the retrieval error

Proof of Proposition 6.11. The state loss $X_k = \sum_b v_b\ell_{k,b}$ lies in $[0, L\omega_{\max}]$ and has $\operatorname{Var} X_k \le \mathbb{E}X_k^2 \le L^2\mathbb{E}[\omega^2] = L^2\mathbb{E}[\omega]^2 m/m_{\text{eff}}$. With $x = \log(2K/\delta)$, Bernstein's inequality gives $|\hat{R}_k - R_k| \le \varepsilon$ with $\varepsilon = \sqrt{2x\operatorname{Var} X_k/m} + 2L\omega_{\max}x/(3m)$ except on an event of probability at most δ/K, and a union bound over k makes this hold for every k with probability at least $1 - \delta$. Then $R_{\hat{k}} \le \hat{R}_{\hat{k}} + \varepsilon \le \hat{R}_{k^*} + \varepsilon \le R_{k^*} + 2\varepsilon$ for the best candidate k^*, and 2ε is at most the stated bound. □

Proof of Proposition 6.12. The set $S = \{(F(w), G(w)) + u : \ w \in W,\ u \in [0,\infty)^2\}$ is convex, since F and G are convex and W is convex. Weak optimality says that $(F(w^*), G(w^*)) - (0,\infty)^2$ does not meet S, so the point lies on the boundary of S, and a supporting line gives $\lambda \ne 0$ with $\lambda\cdot s \ge \lambda\cdot(F(w^*), G(w^*))$ for all $s \in S$. Since S is closed upward, $\lambda \ge 0$; normalize λ to $(\alpha, 1-\alpha)$. □

Proof of Lemma 6.13. Write $x = F(w_1) - F(w_2)$ and $y = G(w_2) - G(w_1)$. Optimality of w_1 and of w_2 gives $\alpha_1 x \le (1 - \alpha_1)y$ and $(1 - \alpha_2)y \le \alpha_2 x$. Multiplying the first by $1 - \alpha_2$ and the second by $1 - \alpha_1$ and chaining, $(1 - \alpha_2)\alpha_1 x \le (1 - \alpha_1)\alpha_2 x$, that is $(\alpha_2 - \alpha_1)x \ge 0$, so $x \ge 0$. Then the first inequality gives $(1 - \alpha_1)y \ge \alpha_1 x \ge 0$, so $y \ge 0$. □

References

[1] O. Lamminpää, J. Susiluoto, J. Hobbs, J. McDuffie, A. Braverman and H. Owhadi. Forward model emulator for atmospheric radiative transfer using Gaussian processes and cross validation. *Atmospheric Measurement Techniques*, 18:673–694, 2025.

[2] S. Arlot and A. Celisse. A survey of cross-validation procedures for model selection. *Statistics Surveys*, 4:40–79, 2010.

[3] O. El Balghiti, A. N. Elmachtoub, P. Grigas and A. Tewari. Generalization bounds in the predict-then-optimize framework. *Mathematics of Operations Research*, 48(4):2043–2065, 2023.

[4] J. M. Bates and C. W. J. Granger. The combination of forecasts. *Operational Research Quarterly*, 20(4):451–468, 1969.

[5] A. Behroozi, C. Shen, D. Kifer and K. Lawson. Sensitivity-constrained neural operators for data-efficient forward and inverse modeling of partial differential equation systems. arXiv:2608.29888, 2026.

[6] S. H. Berguin. Jacobian-enhanced neural networks. arXiv:2406.09132, 2024.

[7] A. Berk, P. Conforti, R. Kennett, T. Perkins, F. Hawes and J. van den Bosch. MODTRAN6: a major upgrade of the MODTRAN radiative transfer code. In *Proceedings of SPIE 9088, Algorithms and Technologies for Multispectral, Hyperspectral, and Ultraspectral Imagery XX*, 2014.

[8] L. Breiman. Stacked regressions. *Machine Learning*, 24:49–64, 1996.

[9] N. D. Brenowitz and C. S. Bretherton. Prognostic validation of a neural network unified physics parameterization. *Geophysical Research Letters*, 45:6289–6298, 2018.

[10] P. G. Brodrick, D. R. Thompson, J. E. Fahlen, M. L. Eastwood, C. M. Sarture, S. R. Lundeen, W. Olson-Duvall, N. Carmon and R. O. Green. Generalized radiative transfer emulation for imaging spectroscopy reflectance retrievals. *Remote Sensing of Environment*, 261:112476, 2021.

[11] T. Bui-Thanh, K. Willcox, O. Ghattas and B. van Bloemen Waanders. Goal-oriented, model-constrained optimization for reduction of large-scale systems. *Journal of Computational Physics*, 224(2):880–896, 2007.

[12] J. Byrd and Z. C. Lipton. What is the effect of importance weighting in deep learning? In *Proceedings of the 36th International Conference on Machine Learning*, PMLR volume 97, pages 872–881, 2019.

[13] G. Camps-Valls, J. Verrelst, J. Muñoz-Marí, V. Laparra, F. Mateo-Jiménez, and J. Gómez-Dans. A survey on Gaussian processes for earth-observation data analysis. *IEEE Geoscience and Remote Sensing Magazine*, 4(2):58–78, 2016.

[14] L. Cao, T. O'Leary-Roseberry, P. K. Jha, J. T. Oden and O. Ghattas. Residual-based error correction for neural operator accelerated infinite-dimensional Bayesian inverse problems. *Journal of Computational Physics*, 486:112104, 2023.

[15] L. Cao, T. O'Leary-Roseberry and O. Ghattas. Derivative-informed neural operator acceleration of geometric MCMC for infinite-dimensional Bayesian inverse problems. *Journal of Machine Learning Research*, 26:1–68, 2025.

[16] E. Cleary, A. Garbuno-Inigo, S. Lan, T. Schneider and A. M. Stuart. Calibrate, emulate, sample. *Journal of Computational Physics*, 424:109716, 2021.

[17] B. J. Connor, H. Boesch, G. Toon, B. Sen, C. Miller, and D. Crisp. Orbiting Carbon Observatory: inverse method and prospective error analysis. *Journal of Geophysical Research: Atmospheres*, 113:D05305, 2008.

[18] C. Cortes, M. Mohri, M. Riley and A. Rostamizadeh. Sample selection bias correction theory. In *Algorithmic Learning Theory (ALT 2008)*, Lecture Notes in Computer Science, volume 5254, pages 38–53, 2008.

[19] C. Cortes, Y. Mansour and M. Mohri. Learning bounds for importance weighting. In *Advances in Neural Information Processing Systems 23*, pages 442–450, 2010.

[20] D. Crisp, H. R. Pollock, R. Rosenberg, L. Chapsky, R. A. M. Lee, et al. The on-orbit performance of the Orbiting Carbon Observatory-2 (OCO-2) instrument and its radiometrically calibrated products. *Atmospheric Measurement Techniques*, 10:59–81, 2017.

[21] T. Cui, J. Martin, Y. M. Marzouk, A. Solonen and A. Spantini. Likelihood-informed dimension reduction for nonlinear inverse problems. *Inverse Problems*, 30(11):114015, 2014.

[22] T. Cui, Y. M. Marzouk and K. E. Willcox. Scalable posterior approximations for large-scale Bayesian inverse problems via likelihood-informed parameter and state reduction. *Journal of Computational Physics*, 315:363–387, 2016.

[23] W. M. Czarnecki, S. Osindero, M. Jaderberg, G. Świrszcz and R. Pascanu. Sobolev training for neural networks. In *Advances in Neural Information Processing Systems 30*, 2017.

[24] N. Dadheech, T.-L. He and A. J. Turner. High-resolution greenhouse gas flux inversions using a machine learning surrogate model for atmospheric transport. *Atmospheric Chemistry and Physics*, 25:5159–5174, 2025.

[25] C. W. O'Dell, B. Connor, H. Bösch, D. O'Brien, C. Frankenberg, et al. The ACOS CO_2 retrieval algorithm — part 1: description and validation against synthetic observations. *Atmospheric Measurement Techniques*, 5:99–121, 2012.

[26] P. L. Donti, B. Amos and J. Z. Kolter. Task-based end-to-end model learning in stochastic optimization. In *Advances in Neural Information Processing Systems 30*, pages 5484–5494, 2017.

[27] M. Drusch, U. Del Bello, S. Carlier, et al. Sentinel-2: ESA's optical high-resolution mission for GMES operational services. *Remote Sensing of Environment*, 120:25–36, 2012.

[28] D. Tanré, M. Herman, P. Y. Deschamps and A. de Leffe. Atmospheric modeling for space measurements of ground reflectances, including bidirectional properties. *Applied Optics*, 18(21):3587–3594, 1979.

[29] A. Eldering, C. W. O'Dell, P. O. Wennberg, D. Crisp, M. R. Gunson, et al. The Orbiting Carbon Observatory-2: first 18 months of science data products. *Atmospheric Measurement Techniques*, 10:549–563, 2017.

[30] S. Elfwing, E. Uchibe and K. Doya. Sigmoid-weighted linear units for neural network function approximation in reinforcement learning. *Neural Networks*, 107:3–11, 2018.

[31] A. N. Elmachtoub and P. Grigas. Smart "predict, then optimize". *Management Science*, 68(1):9–26, 2022.

[32] V. Elvira, L. Martino and C. P. Robert. Rethinking the effective sample size. *International Statistical Review*, 90(3):525–550, 2022.

[33] C. Emde, R. Buras-Schnell, A. Kylling, B. Mayer, J. Gasteiger, U. Hamann, J. Kylling, B. Richter, C. Pause, T. Dowling and L. Bugliaro. The libRadtran software package for radiative transfer calculations (version 2.0.1). *Geoscientific Model Development*, 9:1647–1672, 2016.

[34] B.-C. Gao, M. J. Montes, C. O. Davis and A. F. H. Goetz. Atmospheric correction algorithms for hyperspectral remote sensing data of land and ocean. *Remote Sensing of Environment*, 113:S17–S24, 2009.

[35] Z. Gao, L. Yan and T. Zhou. Adaptive operator learning for infinite-dimensional Bayesian inverse problems. *SIAM/ASA Journal on Uncertainty Quantification*, 12(4):1389–1423, 2024.

[36] D. Gogolashvili, M. Zecchin, M. Kanagawa, M. Kountouris and M. Filippone. When is importance weighting correction needed for covariate shift adaptation? arXiv:2303.04020, 2023.

[37] G. H. Golub, M. Heath, and G. Wahba. Generalized cross-validation as a method for choosing a good ridge parameter. *Technometrics*, 21(2):215–223, 1979.

[38] R. Gupta and Q. Zhang. Data-driven decision-focused surrogate modeling. *AIChE Journal*, 70(4):e18338, 2024.

[39] B. Hamzi and H. Owhadi. Learning dynamical systems from data: a simple cross-validation perspective, part I: parametric kernel flows. *Physica D: Nonlinear Phenomena*, 421:132817, 2021.

[40] T. Helin, A. M. Stuart, A. L. Teckentrup and K. C. Zygalakis. Introduction to Gaussian process regression in Bayesian inverse problems, with new results on experimental design for weighted error measures. In *Monte Carlo and Quasi-Monte Carlo Methods (MCQMC 2022)*, Springer Proceedings in Mathematics & Statistics 460, pages 49–79. Springer, Cham, 2024.

[41] M. D. Himes, J. Harrington, A. D. Cobb, et al. Accurate machine-learning atmospheric retrieval via a neural-network surrogate model for radiative transfer. *The Planetary Science Journal*, 3(4):91, 2022.

[42] J. Huang, A. Gretton, K. M. Borgwardt, B. Schölkopf and A. J. Smola. Correcting sample selection bias by unlabeled data. In *Advances in Neural Information Processing Systems 19*, 2007.

[43] E. L. Ionides. Truncated importance sampling. *Journal of Computational and Graphical Statistics*, 17(2):295–311, 2008.

[44] J. Kaipio and E. Somersalo. *Statistical and Computational Inverse Problems*. Springer, New York, 2005.

[45] J. Kaipio and E. Somersalo. Statistical inverse problems: discretization, model reduction and inverse crimes. *Journal of Computational and Applied Mathematics*, 198(2):493–504, 2007.

[46] M. C. Kennedy and A. O'Hagan. Predicting the output from a complex computer code when fast approximations are available. *Biometrika*, 87(1):1–13, 2000.

[47] D. P. Kingma and J. Ba. Adam: a method for stochastic optimization. In *International Conference on Learning Representations*, 2015.

[48] L. Kish. *Survey Sampling*. Wiley, New York, 1965.

[49] A. N. Kolmogorov. On the representation of continuous functions of several variables by superpositions of continuous functions of one variable and addition. *Doklady Akademii Nauk SSSR*, 114:953–956, 1957.

[50] P. Kopparla, V. Natraj, R. Spurr, R.-L. Shia, D. Crisp and Y. L. Yung. A fast and accurate PCA based radiative transfer model: extension to the broadband shortwave region. *Journal of Quantitative Spectroscopy and Radiative Transfer*, 173:65–71, 2016.

[51] S. Y. Kotchenova, E. F. Vermote, R. Matarrese and F. J. Klemm, Jr. Validation of a vector version of the 6S radiative transfer code for atmospheric correction of satellite data. Part I: path radiance. *Applied Optics*, 45(26):6762–6774, 2006.

[52] S. Kpotufe and G. Martinet. Marginal singularity and the benefits of labels in covariate-shift. *Annals of Statistics*, 49(6):3299–3323, 2021.

[53] V. M. Krasnopolsky, M. S. Fox-Rabinovitz, and D. V. Chalikov. New approach to calculation of atmospheric model physics: accurate and fast neural network emulation of longwave radiation in a climate model. *Monthly Weather Review*, 133:1370–1383, 2005.

[54] Y. Kurniawan, T. B. Neilsen, B. L. Francis, et al. An information-matching approach to optimal experimental design and active learning. *Applied Physics Letters*, 128(6):064104, 2026.

[55] T. O'Leary-Roseberry, U. Villa, P. Chen and O. Ghattas. Derivative-informed projected neural networks for high-dimensional parametric maps governed by PDEs. *Computer Methods in Applied Mechanics and Engineering*, 388:114199, 2022.

[56] T. O'Leary-Roseberry, P. Chen, U. Villa and O. Ghattas. Derivative-informed neural operator: an efficient framework for high-dimensional parametric derivative learning. *Journal of Computational Physics*, 496:112555, 2024.

[57] J. Li and Y. M. Marzouk. Adaptive construction of surrogates for the Bayesian solution of inverse problems. *SIAM Journal on Scientific Computing*, 36(3):A1163–A1186, 2014.

[58] C. Lieberman and K. Willcox. Goal-oriented inference: approach, linear theory, and application to advection diffusion. *SIAM Journal on Scientific Computing*, 34(4):A1880–A1904, 2012.

[59] C. Lieberman and K. Willcox. Nonlinear goal-oriented Bayesian inference: application to carbon capture and storage. *SIAM Journal on Scientific Computing*, 36(3):B427–B449, 2014.

[60] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y. Hou, and M. Tegmark. KAN: Kolmogorov–Arnold networks. arXiv:2404.19756, 2024.

[61] C. Ma, R. Pathak and M. J. Wainwright. Optimally tackling covariate shift in RKHS-based nonparametric regression. *The Annals of Statistics*, 51(2):738–761, 2023.

[62] J. Mandi, J. Kotary, S. Berden, M. Mulamba, V. Bucarey, T. Guns and F. Fioretto. Decision-focused learning: foundations, state of the art, benchmark and future opportunities. *Journal of Artificial Intelligence Research*, 80:1623–1701, 2024.

[63] S. A. Mattis and B. Wohlmuth. Goal-oriented adaptive surrogate construction for stochastic inversion. *Computer Methods in Applied Mechanics and Engineering*, 339:36–60, 2018.

[64] B. Mayer and A. Kylling. Technical note: the libRadtran software package for radiative transfer calculations — description and examples of use. *Atmospheric Chemistry and Physics*, 5:1855–1877, 2005.

[65] M. A. A. Mazid and N. Rishe. Multi-fidelity emulation of atmospheric correction coefficients with physics-guided Kolmogorov–Arnold networks. *Remote Sensing*, 18(11):1826, 2026.

[66] J. Quiñonero-Candela, M. Sugiyama, A. Schwaighofer and N. D. Lawrence (eds.). *Dataset Shift in Machine Learning*. MIT Press, Cambridge, MA, 2008.

[67] J. L. Gómez-Dans, P. E. Lewis, and M. Disney. Efficient emulation of radiative transfer codes using Gaussian processes and application to land surface parameter inferences. *Remote Sensing*, 8(2):119, 2016.

[68] H. Owhadi and G. R. Yoo. Kernel flows: from learning kernels from data into the abyss. *Journal of Computational Physics*, 389:22–47, 2019.

[69] A. Paszke, S. Gross, F. Massa, et al. PyTorch: an imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32*, 2019.

[70] B. Peherstorfer, K. Willcox, and M. Gunzburger. Survey of multifidelity methods in uncertainty propagation, inference, and optimization. *SIAM Review*, 60(3):550–591, 2018.

[71] P. Perdikaris, M. Raissi, A. Damianou, N. D. Lawrence, and G. E. Karniadakis. Nonlinear information fusion algorithms for data-efficient multi-fidelity modelling. *Proceedings of the Royal Society A*, 473:20160751, 2017.

[72] L. Prechelt. Early stopping — but when? In *Neural Networks: Tricks of the Trade*, volume 1524 of *Lecture Notes in Computer Science*, pages 55–69. Springer, 1998.

[73] C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

[74] S. Rasp, M. S. Pritchard, and P. Gentine. Deep learning to represent subgrid processes in climate models. *Proceedings of the National Academy of Sciences*, 115(39):9684–9689, 2018.

[75] P. Reiser, J. E. Aguilar, A. Guthke and P.-C. Bürkner. Uncertainty quantification and propagation in surrogate-based Bayesian inference. *Statistics and Computing*, 35(3):66, 2025.

[76] R. Richter and D. Schläpfer. Geo-atmospheric processing of airborne imaging spectrometry data. Part 2: atmospheric/topographic correction. *International Journal of Remote Sensing*, 23(13):2631–2649, 2002.

[77] J. P. Rivera, J. Verrelst, J. Gómez-Dans, J. Muñoz-Marí, J. Moreno, and G. Camps-Valls. An emulator toolbox to approximate radiative transfer models with statistical learning. *Remote Sensing*, 7(7):9347–9370, 2015.

[78] C. D. Rodgers. *Inverse Methods for Atmospheric Sounding: Theory and Practice*. World Scientific, 2000.

[79] J. Sacks, W. J. Welch, T. J. Mitchell, and H. P. Wynn. Design and analysis of computer experiments. *Statistical Science*, 4(4):409–423, 1989.

[80] P. Semler and M. Weiser. Adaptive gradient enhanced Gaussian process surrogates for inverse problems. arXiv:2404.01864, 2024.

[81] J. Vicent Servera, J. P. Rivera-Caicedo, J. Verrelst, J. Muñoz-Marí, N. Sabater, B. Berthelot, G. Camps-Valls and J. Moreno. Systematic assessment of MODTRAN emulators for atmospheric correction. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–17, 2022.

[82] C. Sgattoni, L. Sgheri and M. Chung. A physics-aware data-driven surrogate approach for fast atmospheric radiative transfer inversion. arXiv:2410.22609, 2024.

[83] S. Shah, K. Wang, B. Wilder, A. Perrault and M. Tambe. Decision-focused learning without decision-making: learning locally optimized decision losses. In *Advances in Neural Information Processing Systems 35*, pages 1320–1332, 2022.

[84] H. Shimodaira. Improving predictive inference under covariate shift by weighting the log-likelihood function. *Journal of Statistical Planning and Inference*, 90(2):227–244, 2000.

[85] Y. Shmalo. Learned representations and kernel corrections for radiative-transfer emulation. Preprint, 2026.

[86] L. N. Smith and N. Topin. Super-convergence: very fast training of neural networks using large learning rates. *Proc. SPIE 11006, Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications*, 1100612, 2019.

[87] P. Somkuti, H. Bösch, V. Natraj and P. Kopparla. Application of a PCA-based fast radiative transfer model to XCO2 retrievals in the shortwave infrared. *Journal of Geophysical Research: Atmospheres*, 122(19):10477–10496, 2017.

[88] P. Somkuti, H. Bösch and R. J. Parker. The significance of fast radiative transfer for hyperspectral SWIR XCO2 retrievals. *Atmosphere*, 11(11):1219, 2020.

[89] A. Spantini, T. Cui, K. Willcox, L. Tenorio and Y. Marzouk. Goal-oriented optimal approximations of Bayesian linear inverse problems. *SIAM Journal on Scientific Computing*, 39:S167–S196, 2017.

[90] M. L. Stein. *Interpolation of Spatial Data: Some Theory for Kriging*. Springer, 1999.

[91] M. Stone. Cross-validatory choice and assessment of statistical predictions. *Journal of the Royal Statistical Society: Series B*, 36(2):111–147, 1974.

[92] A. M. Stuart. Inverse problems: a Bayesian perspective. *Acta Numerica*, 19:451–559, 2010.

[93] A. M. Stuart and A. L. Teckentrup. Posterior consistency for Gaussian process approximations of Bayesian posterior distributions. *Mathematics of Computation*, 87(310):721–753, 2018.

[94] M. Sugiyama, M. Krauledat and K.-R. Müller. Covariate shift adaptation by importance weighted cross validation. *Journal of Machine Learning Research*, 8:985–1005, 2007.

[95] M. Sugiyama and M. Kawanabe. *Machine Learning in Non-Stationary Environments: Introduction to Covariate Shift Adaptation*. MIT Press, Cambridge, MA, 2012.

[96] D. R. Thompson, V. Natraj, R. O. Green, M. C. Helmlinger, B.-C. Gao and M. L. Eastwood. Optimal estimation for imaging spectrometer atmospheric correction. *Remote Sensing of Environment*, 216:355–373, 2018.

[97] D. R. Thompson, A. Braverman, P. G. Brodrick, et al. Quantifying uncertainty for remote spectroscopy of surface composition. *Remote Sensing of Environment*, 247:111898, 2020.

[98] P. Ukkonen. Exploring pathways to more accurate machine learning emulation of atmospheric radiative transfer. *Journal of Advances in Modeling Earth Systems*, 14:e2021MS002875, 2022.

[99] M. A. Veerman, R. Pincus, R. Stoffer, C. M. van Leeuwen, D. Podareanu, and C. C. van Heerwaarden. Predicting atmospheric optical properties for radiative transfer computations using neural networks. *Philosophical Transactions of the Royal Society A*, 379:20200095, 2021.

[100] A. Vehtari, D. Simpson, A. Gelman, Y. Yao and J. Gabry. Pareto smoothed importance sampling. *Journal of Machine Learning Research*, 25(72):1–58, 2024.

[101] E. F. Vermote, D. Tanré, J. L. Deuzé, M. Herman and J.-J. Morcrette. Second simulation of the satellite signal in the solar spectrum, 6S: an overview. *IEEE Transactions on Geoscience and Remote Sensing*, 35(3):675–686, 1997.

[102] J. Verrelst, N. Sabater, J. P. Rivera, J. Muñoz-Marí, J. Vicent, G. Camps-Valls and J. Moreno. Emulation of leaf, canopy and atmosphere radiative transfer models for fast global sensitivity analysis. *Remote Sensing*, 8(8):673, 2016.

[103] J. Verrelst, M. Morata, J. L. García-Soria, Y. Sun, J. Qi and J. P. Rivera-Caicedo. RTM surrogate modeling in optical remote sensing: a review of emulation for vegetation and atmosphere applications. *Remote Sensing*, 17(21):3618, 2025.

[104] B. Wilder, B. Dilkina and M. Tambe. Melding the data-decisions pipeline: decision-focused learning for combinatorial optimization. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33(1):1658–1665, 2019.

[105] C. K. I. Williams and M. Seeger. Using the Nyström method to speed up kernel machines. In *Advances in Neural Information Processing Systems 13*, pages 682–688. MIT Press, 2001.

[106] D. H. Wolpert. Stacked generalization. *Neural Networks*, 5(2):241–259, 1992.

[107] O. Zahm, T. Cui, K. Law, A. Spantini and Y. Marzouk. Certified dimension reduction in nonlinear Bayesian inverse problems. *Mathematics of Computation*, 91(336):1789–1835, 2022.

[108] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, `https://github.com/yspennstate/math-brain-public`.

[109] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, `https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24`.

[110] Y. Shmalo, *Brain Framework*, public software repository, 2026, `https://github.com/yspennstate/brain-framework-public`.